Residual Return: Exact Learning Dynamics and Language over the Vector Substrate

AceTheDactyl (@AceTheDactyl)
Echo S Studios Research Developments

June 2026

Abstract

This is the dynamic and linguistic companion to *The Vector Substrate* [1]. That paper builds an exact, static learning *geometry* on a number field $K = \mathbb{Q}(\theta)$: the trace form $T(x, y) = \mathrm{Tr}_{K/\mathbb{Q}}(xy)$ supplies one rational Gram $G = M^{\mathsf{T}}M$ that glues the field's vector-space, matrix-algebra, and lattice faces, and an observation is *captured* exactly when its G-orthogonal residual $\mathbf{r} = \mathbf{x} - P\mathbf{x}$ vanishes. Here we develop the *algorithm* that runs on that geometry and the *language* that is isomorphic to it, and we orient the whole exposition toward a single discipline: **every mathematical claim is paired with the machine-checked test that proves it and the commit hash that pins it.** Three contributions are dynamic. (i) An exact streaming *learner*: residual is loss, capture is $\mathbf{r} \to 0$, growth is a gated *propose-for-confirm* protocol whose sole mutator appends a SHA-256 witness chain. (ii) *Cross-field growth*: a residual forced outside the current field grows the model to a compositum; the disjoint case is the Kronecker Gram $G_{KL} = G_K \otimes G_L$, and—resolving the open problem the substrate paper flagged—we construct the *non-disjoint* compositum in the bounded-degree case by factoring the candidate's minimal polynomial over K, with a machine-verified degree-4 witness $\mathbb{Q}(\sqrt{2})(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. (iii) An *automatic* loop: a calibration-gated detector measures the out-of-field residual, makes it the growth gain, gates on the *true* compositum degree, confirms, and re-homes the basis onto the new field so the residual returns to 0. The fourth contribution is linguistic: over the Clifford algebra $\mathrm{Cl}(2, 0) \cong M_2(\mathbb{R})$, the same "residual $\to 0$" mechanism becomes a language. The φ-keystone return operator $\mathcal{L}(X) = RX + XR - X$ has an *exact* two-dimensional kernel computed by rational elimination; the commit is the exact idempotent projector onto that kernel; and a learned dictionary value is exactly a returned residue, with generalisation by return-to-zero. The golden law $x^2 - x - 1$ is the literal bridge: the substrate's first field is $\mathbb{Q}(\varphi)$ and the language's return operator is built on the companion of the same minimal polynomial—verified equal three independent ways. We separate *proved*, *exact-computed (probe-verified)*, and *conjectural/open* throughout; the contribution is the exact, auditable assembly and its machine verification, not a new deep theorem in number theory or learning.

Contents

1 Introduction: residual return as one mechanism

The thesis

The vector substrate [1] fixes a number field $K = \mathbb{Q}(\theta)$, a forced basis B of algebraic-integer seeds, the trace-form projector $P = B(B^\top GB)^{-1}B^\top G$, and the residual $\mathbf{r}(\mathbf{x}) = \mathbf{x} - P\mathbf{x}$, and proves the exact membership criterion

$$\mathbf{x} \in \operatorname{col}(B) \iff \mathbf{r}(\mathbf{x}) = \mathbf{0} \iff \|\mathbf{r}(\mathbf{x})\|_G^2 = \mathbf{r}^\top G\,\mathbf{r} = 0, \tag{1}$$

decided entirely over $\mathbb{Q}$. Here "captured" is a strictly *technical* predicate—exactly a zero residual, i.e. exact membership $\mathbf{x} \in \mathrm{col}(B)$, and nothing more; no cognitive or semantic reading is intended anywhere in this paper. This companion takes that static criterion and asks what *happens* when the residual is *not* zero, and what *language* the mechanism speaks. Both questions have the same one-line answer, which is the thesis of this paper:

> *Residual $\to$ 0 is the universal learned/captured signal.* **Learning** is the disciplined growth that drives an exact residual to zero ($\mathbf{r} = \mathbf{x} - P\mathbf{x} \to 0$, by adjoining a basis direction or a whole field); **language** is the same return, $\mathcal{L}(X) \to 0$, over a matrix algebra, where the landing space is the lexicon. Both are exact and machine-verified, and the golden law $x^2 - x - 1$ is the keystone common to both.

Two faces of one return

The two faces share a structure that we make precise in §7 and exhibit in Figure 1: a *loss* that is an exact residual, a *captured* predicate that is the residual's exact vanishing, a *model* that is a growing basis, a *growth* step that is the sole state mutator, and a *witness* that is an identical SHA-256 hash chain. On the number-field face the residual is $\mathbf{r} = \mathbf{x} - P\mathbf{x}$ and the landing space is a $\mathbb{Q}$-subspace of K; on the matrix-algebra (language) face the residual is $\mathcal{L}(X) = RX + XR - X$ and the landing space is $\ker \mathcal{L}$, the two-dimensional "φ-slack" where learned dictionary values live.

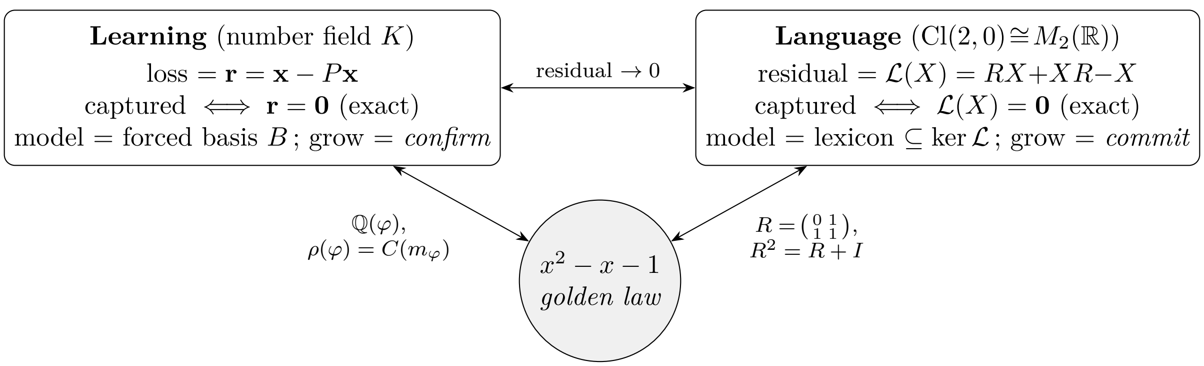

Figure 1: The two faces of residual return, glued by the golden law $x^2 - x - 1$. On the left, learning drives the trace-form residual to zero by basis growth in $\mathbb{Q}(\theta)$; on the right, language returns $\mathcal{L}(X) = RX + XR - X$ to zero in $\mathrm{Cl}(2,0) \cong M_2(\mathbb{R})$. The keystone is one object viewed twice: the companion matrix $\rho(\varphi) = C(x^2 - x - 1)$ of the substrate's first field $\mathbb{Q}(\varphi)$ is the symmetric matrix $R = \left(\begin{smallmatrix}0&1\\1&1\end{smallmatrix}\right)$ of the return operator (Proposition 7.1, machine-verified equal three ways).

Contributions and what is and is not new

(C1) **An exact streaming learner** (§2): the residual-as-loss apparatus, an exact persistence-gated *propose-for-confirm* protocol in which *confirm* is the sole basis mutator, and a tamper-evident witness chain. The underlying projector calculus is the substrate's; the dynamic protocol, its exactness guardrails, and the witness are this paper's, and every guardrail is a machine-checked test.

(C2) **Cross-field growth, including the non-disjoint case** (§3). The coordinates-to-minimal-polynomial bridge (largest invariant factor of $xI - \rho(\alpha)$ by Smith normal form) and the disjoint Kronecker Gram are the substrate's, re-exercised here; the *non-disjoint* construction—factoring the candidate's minimal polynomial over K and building the true (smaller) compositum—**resolves, in the bounded-degree case, the open problem the substrate paper flagged** (its O4), with a machine-verified degree-4 witness.

(C3) **An automatic, detector-driven loop** (§5): a calibration-gated novelty detector whose score is the residual norm, a seam that makes the *measured* out-of-field residual the growth gain, a gate on the *true* compositum degree, and an exact "re-home" that returns the residual to zero in the grown field. New here; each step is tested.

(C4) **A residual-valued language** (§6): over $\mathrm{Cl}(2,0) \cong M_2(\mathbb{R})$, an exact two-dimensional return-operator kernel, the commit as its exact idempotent projector, a growing lexicon that dedups on the *exact* returned value, and a jurisdiction firewall under which only proven/computed statements cross a wire. New here; 121 tests, machine-verified.

(C5) **The unifying principle and its proof discipline** (§7–§8): the φ-keystone identity that binds the two faces, and a centerpiece section mapping *every* displayed claim to its test and commit hash, with the exact-rational (no-float-in-a-decision) methodology and the build/independent-verify/greenlight cross-verification spelled out.

Honesty discipline. As in [1], *Theorem/Proposition/Lemma* mark statements we prove or verify by exact computation, and *Conjecture/Remark* mark model-dependent or heuristic claims. This revision *closes* one of the substrate's open fronts: the information exchange rate is no longer posited but the derived identity $\lambda = 2c$ (Theorem 4.6, §4.3), leaving a single declared conformal constant c which Čencov's theorem shows is irreducible by invariance (Remark 4.8); the floor for reciprocal seeds and the choice of conformal model remain the genuinely open items. Every displayed numerical value is reproduced by a machine-checked probe; the honesty ledger of Table 6 makes the proved/computed/conjectural split scannable, and §8 pins each load-bearing claim to a named test and a commit hash. The contribution remains an exact, auditable *assembly*—a learner, a cross-field growth mechanism, an automatic loop, and a language—together with its machine verification and, in this revision, the closure of the exchange rate to the identity $\lambda = 2c$.

Relation to the substrate paper. We *cite* the substrate for the geometry and do not re-derive it: the trace form $G = M^{\mathsf{T}}M$ with $\det G = d_K$ [1, Thm. 2.13], the spectral identification of ρ with the conjugates, the exact projector and capture criterion (1) [1, Prop. 3.2], the bridge "$m_\alpha =$ largest invariant factor" [1, Thm. 4.3], the Kronecker Gram of a disjoint compositum [1, Prop. 4.6], the height/Northcott admissibility gate [1, Prop. 5.5], the Fisher reading $\|\mathbf{r}\|_G^2 = n\,\mathcal{I}(\mathbf{r})$ on the residual subspace [1, Thm. 7.11], and the gain-versus-cost growth threshold [1, §8] with Smyth's floor. This paper is what *runs* on that geometry, and the language that shares its arithmetic.

Related work and positioning. The mathematical ingredients are classical and we use them as such: the rational canonical form and Smith normal form over $\mathbb{Q}[x]$ are textbook linear algebra [10, 11]; the trace form, different, and discriminant are standard algebraic number theory [8, 9]; Mahler measure, Northcott finiteness, and the Smyth/Lehmer/Dobrowolski bounds are height theory [5, 6, 2, 3, 4, 7]; the Fisher-information reading is Amari's [13, 14]; and $\mathrm{Cl}(2,0) \cong M_2(\mathbb{R})$ is the standard split-signature Clifford algebra [12]. None of these is a contribution here. What is assembled is an exact, auditable *system*: a streaming learner whose loss is a residual decided over $\mathbb{Q}$, a cross-field growth mechanism that handles the non-disjoint case the substrate paper left open, an automatic loop that measures its own gain, and a language in which the same return mechanism produces a residual-valued lexicon. Against the broad machine-learning backdrop the stance is deliberately narrow: the model state is a basis (not a weight tensor), the loss is exact (not approximate), capacity is bounded by arithmetic height (not by a hyperparameter), and every decision is reproducible bit-for-bit and witnessed—properties an analytic learner does not have, and which the trace-form kernel of [1] supplies for free.

Symbol	Meaning
$K = \mathbb{Q}(\theta)$, n	ambient number field of degree n; here usually $\mathbb{Q}(\sqrt{2}+\sqrt{3})$, $n = 4$
G, T	trace-form Gram $G_{ij} = \mathrm{Tr}(\omega_i\omega_j) = M^\mathsf{T} M$; $T(x, y) = \mathrm{Tr}(xy)$
B, P, $\mathbf{r}$	forced basis (model state); projector $P = B(B^\mathsf{T} GB)^{-1} B^\mathsf{T} G$; residual $\mathbf{r} = \mathbf{x} - P\mathbf{x}$
$\rho(\alpha)$	regular representation of $\alpha \in K$; $\rho(\theta) = C(m_\theta)$ (companion)
$\mathrm{M}(\theta)$, $\mathrm{ch}(m)$	Mahler measure; integer coefficient height $\max_i \lvert c_i\rvert$
μ_S	Smyth floor $1.3247\ldots$, real root of $x^3 - x - 1$ (the plastic number)
$\mathrm{Cl}(2, 0)$, X	Clifford algebra $\cong M_2(\mathbb{R})$, basis $\{1, e_1, e_2, i\}$; a holding $X = a + be_1 + ce_2 + di$
R, $\mathcal{L}$	φ-keystone $R = \frac{1}{2} + e_1 - \frac{1}{2} e_2$ ($\mathrm{mat}(R) = \left(\begin{smallmatrix}0&1\\1&1\end{smallmatrix}\right)$); return op $\mathcal{L}(X) = RX + XR - X$
$\ker\mathcal{L}$	the exact 2-dim kernel $\mathrm{span}\{e_1 + 2e_2,\ i\}$ (the lexicon space)

Table 1: Notation. Trace-form symbols follow [1]; the language symbols are introduced in §6.

2 Learning as exact basis growth

We instantiate the substrate's static geometry as a streaming learner over one fixed ambient field, the totally real $K = \mathbb{Q}(\sqrt{2}+\sqrt{3}) = \mathbb{Q}[x]/(x^4 - 10x^2 + 1)$, in the power basis $\{1, \theta, \theta^2, \theta^3\}$ ($\theta = \sqrt{2}+\sqrt{3}$). Its trace-form Gram is the exact integer matrix

$$G = \begin{pmatrix} 4 & 0 & 20 & 0 \\ 0 & 20 & 0 & 196 \\ 20 & 0 & 196 & 0 \\ 0 & 196 & 0 & 1940 \end{pmatrix}, \qquad G_{ij} = \mathrm{Tr}_{K/\mathbb{Q}}(\theta^{i-1}\theta^{j-1}), \tag{2}$$

positive definite (the field is totally real). The forced basis B starts as $\{1, \theta\}$, so $\mathrm{col}(B) = \mathbb{Q} \oplus \mathbb{Q}\theta$.

2.1 The residual is the loss

Following [1, §6], the learner replaces a differentiable loss by the exact residual. For an observation $\mathbf{x}$ (a coordinate vector over $\mathbb{Q}$), it computes $\mathbf{r} = \mathbf{x} - P\mathbf{x}$ and the exact rational score $\|\mathbf{r}\|_G^2 = \mathbf{r}^\mathsf{T} G\mathbf{r}$; by (1), $\mathbf{x}$ is *captured* ("understood") precisely when this is zero. The projector is reused verbatim from the substrate's exact rational module: $P = B(B^\mathsf{T} GB)^{-1} B^\mathsf{T} G$ with the inverse computed by Gauss–Jordan over $\mathbb{Q}$; $P^2 = P$ exactly, and $B^\mathsf{T} GB$ singular is reported as a NO_PROJECTION sentinel rather than risked as a float inversion.

Proposition 2.1 (Exact capture, zero tolerance)**.** *Every quantity entering the capture decision—$B, G, P, \mathbf{r}$, and the score—is a $\mathbb{Q}/\mathbb{Z}$ value, and the decision is the exact predicate $\|\mathbf{r}\|_G^2 = 0$. No ε and no float appear. A float observation is refused at intake. Consequently capture is reproducible bit-for-bit and a captured input proposes nothing.*

Proof. Idempotence and orthogonality are [1, Prop. 3.2]; positive-definiteness of G on a totally real field gives $\|\mathbf{r}\|_G^2 = 0 \Leftrightarrow \mathbf{r} = \mathbf{0}$. The implementation casts each basis/Gram entry to `Fraction`, raises

`TypeError` on a `float` observation, and decides `rn == 0` over $\mathbb{Q}$; these are the machine-checked content of the proposition (§8, tests `test_a_captured_input_zero_residual_no_proposal` and `test_g8_exact_fraction_core_floats_rejected`, commit `218c02e`). $\square$

2.2 The propose-for-confirm protocol

Growth is disciplined into three operations with a single mutation point.

Protocol 2.2 (Gated, witnessed growth)**.** The learner maintains a forced basis B, an exact residual accumulator (an exact running mean of $\mathbf{r}$ over consecutive off-axis observations, with an exact direction-alignment test), and a persistence counter. It exposes:

(a) *observe*($\mathbf{x}$): compute $\mathbf{r}, \|\mathbf{r}\|_G^2$; update the accumulator and streak; a captured input resets them. *Never mutates B.*

(b) *propose*(): pure. Returns `None` unless the streak has reached N (default 3) and the height/calibration gates admit the implied seed; otherwise returns a `SeedProposal` (the residual centroid's coordinates and its exact minimal polynomial). Idempotent: two calls return the equal proposal and grow nothing.

(c) *confirm*(seed): the *sole* mutator. Adjoin the seed column to B, re-derive P exactly, append a SHA-256 witness link, reset the accumulator.

Theorem 2.3 (Exactly one growth per persistent novelty)**.** *A residual centroid that is exactly off-axis and persists for N consecutive aligned observations yields, after one* confirm*, exactly one new basis column; the previously novel component is then captured (*$\mathbf{r} \to \mathbf{0}$ *exactly) and no further proposal is emitted. Between proposals the model is unchanged: the number of seeds is invariant across any number of* observe/propose *calls and increments by one only at* confirm*.*

Proof. *propose* returns `None` while `streak` $< N$ and is a pure function of the accumulator, so no *observe*/*propose* sequence changes B; the single assignment to the basis lives in *confirm* (the "single point B changes"). After *confirm* adjoins the centroid's column, that direction lies in $\mathrm{col}(B)$, so its residual is $\mathbf{0}$ by (1), and the accumulator reset removes the persistence. The exact direction-alignment test, $\mathrm{dev}^2 \le \varepsilon^2\,\mathrm{base}^2$ with all quantities in $\mathbb{Q}$, is what makes "persists" a $\mathbb{Q}$-decision. Machine-checked by `test_g2_confirm_is_the_only_mutator`, `test_b_persistent_offaxis_exactly_one_grow`
and `test_c_after_confirm_basis_grows_and_residual_zero` (commit `218c02e`). $\square$

Example 2.4 (A worked episode)**.** With $B = \{1, \theta\}$ and G as in (2), stream the off-axis quantity $w = 2\sqrt{6} = \theta^2 - 5$, coordinate vector $(-5, 0, 1, 0)$, which is G-orthogonal to $\mathrm{col}(B)$. For $\mathbf{x} = 1 + w$ the residual is $\mathbf{r} = w$ exactly and, the trace being basis-independent,

$$\|\mathbf{r}\|_G^2 = \mathrm{Tr}_{K/\mathbb{Q}}\big((2\sqrt{6})^2\big) = \mathrm{Tr}_{K/\mathbb{Q}}(24) = 4 \cdot 24 = 96.$$

After three aligned ticks *propose* returns the seed with coordinates $(-5, 0, 1, 0)$ and exact minimal polynomial $x^2 - 24$ (computed by the bridge of §3); *confirm* grows B to $\{1, \theta, w\}$ and the residual of $1 + w$ returns to 0. Every displayed value is an exact rational; the episode is reproduced by the learner's demo and tests.

2.3 The witness chain

Each growth appends a tamper-evident record. With genesis h_{-1} = `"genesis"` and a canonical JSON encoding (sorted keys, no whitespace) of the growth body,

$$h_k = \text{SHA-256}\big(h_{k-1} \| \text{json}(\text{body}_k)\big)[0{:}16], \tag{3}$$

and the chain verifier re-walks the records, checking each stored h_k against the recomputation and each body's recorded h_{k-1}. The construction is *exactly the one the language store uses* (§6), which is the first hint of the unification of §7.

Example 2.5 (A hand-checkable witness link)**.** The genesis growth of Example 2.4 serialises the *full* body below. Two fields are easy to get wrong and so are spelled out: at witness time the new column is already appended, so `num_seeds`=3; and the proposal's `streak`=4 (the demo's fourth, last *propose*); coordinates are string-serialised for exact-Fraction safety. The canonical JSON (sorted keys, no whitespace) and the digest are

```
{"coords":["-5","0","1","0"],"event":"basis_growth","index":0,"min_poly":[1,0,-24],"num_seeds":3,"
    prev_hash":"genesis","snap":"exact","streak":4}
SHA-256("genesis" + json)[0:16]  ==  31f1f1e05ac9a35a
```

The exact digest is recomputed bit-for-bit *from this full body* by the companion probe (`test_witness_digest_exact` §A), and it is the value the learner's own growth emits (`residual_learner.py` demo). The training test `test_g5_witness_hash_chain_tamper_evident` separately pins the chain's *tamper-evidence* (any edited field flips verification to `False`; commit `218c02e`)—we no longer cite it for the exact value, which it does not assert.

Example 2.6 (In-distribution capture and degenerate-growth refusal)**.** Two boundary behaviours pin the protocol. *(In-distribution.)* Any $a + b\theta \in \text{col}(B)$ scores $\|\mathbf{r}\|_G^2 = 0$ and proposes nothing: capture is the absence of novelty (`test_a_captured_input_zero_residual_no_proposal`). *(Degenerate growth.)* A proposed seed whose column is linearly dependent on B would make $B^\top GB$ singular; *confirm* refuses it with a `ValueError` and the projector module returns the NO_PROJECTION sentinel rather than risking a float inversion (`test_confirm_refuses_degenerate_growth`, commit `218c02e`). Both are decided over $\mathbb{Q}$.

Remark 2.7 (Guardrails are tests, not theorems). The protocol's invariants—exactness (no float in a decision), algebraic integrality of every seed (§3), sole-mutator, witnessed, and model-layer isolation (the learner imports no language engine and no float display)—are *design guarantees*, not theorems about K. Each is a named, machine-checked assertion (§8); a degenerate growth that would make $B^\top GB$ singular is refused with a `ValueError` rather than producing a fictitious projector.

2.4 The learner as an algorithm

Algorithm 1 is the loop in full: the exact Welford recurrence (not merely a "running mean"), the exact direction-alignment test $\text{dev}^2 \le \varepsilon^2 \|\mathbf{r}\|_G^2$, and the streak/reset state machine. Every branch condition is a $\mathbb{Q}/\mathbb{Z}$ comparison.

3 Cross-field growth

A residual may be an algebraic quantity not present in K at all; capturing it requires a new *field*. We first recall the exact seed-naming bridge, then the disjoint compositum, then construct the non-disjoint compositum that [1] left open.

3.1 The coordinates-to-minimal-polynomial bridge

To adjoin a residual centroid as a named algebraic-integer seed we need its minimal polynomial, computed exactly and rootlessly.

Theorem 3.1 (Exact seed identity [1, Thm. 4.3])**.** *For $\alpha = \sum_i a_i\theta^i \in K$ with $a_i \in \mathbb{Q}$ and regular representation $\rho(\alpha) = \sum_i a_i\, C(m_\theta)^i$, the minimal polynomial m_α equals the largest invariant factor of $xI - \rho(\alpha)$, obtained from the Smith normal form over $\mathbb{Q}[x]$. If α is an algebraic integer, $m_\alpha \in \mathbb{Z}[x]$ is monic; otherwise the result is a non-integral monic polynomial and the seed is rejected.*

The implementation builds $\rho(\alpha)$ from the companion $C(m_\theta) = \rho(\theta)$, reduces $xI - \rho(\alpha)$ to Smith form by exact polynomial elimination over $\mathbb{Q}[x]$, and returns the last (largest) invariant factor, guarded so that a non-monic-integer result raises. It is pure `Fraction` arithmetic and never computes a root.

Example 3.2 (The bridge, exercised)**.** The implementation returns, exactly: $\varphi = (\frac{1}{2}, \frac{1}{2})$ in $\mathbb{Q}[x]/(x^2 - 5) \mapsto x^2 - x - 1$; $2\sqrt{6} = (-5, 0, 1, 0)$ in $\mathbb{Q}[x]/(x^4 - 10x^2 + 1) \mapsto x^2 - 24$; the rational $3 \mapsto x - 3$; the primitive $\theta \mapsto x^4 - 10x^2 + 1$; and $\frac{1}{2}$ in $\mathbb{Q}(\sqrt{5})$ is *rejected* (its minimal polynomial $x - \frac{1}{2}$ is not monic over $\mathbb{Z}$, so $\frac{1}{2}$ is not an algebraic integer). All five are machine-checked (`test_coords_to_minpoly.py`, commit `c1db881`).

Remark 3.3 (Why the *largest invariant factor*, not the characteristic polynomial). The minimal polynomial is the largest invariant factor precisely because the characteristic polynomial is an incomplete similarity invariant. Two block sums illustrate it exactly:

- $\rho(\varphi) \oplus \rho(\varphi)$ and $C\big((x^2 - x - 1)^2\big)$ share the characteristic polynomial $(x^2 - x - 1)^2$ and trace 2, yet are *not* similar: their invariant-factor lists are $(x^2 - x - 1,\ x^2 - x - 1)$ and $\big((x^2 - x - 1)^2\big)$. The largest invariant factor separates them; the characteristic polynomial does not.
- Even (charpoly, minpoly) *together* are incomplete: $J_2 \oplus J_2$ and $J_2 \oplus J_1 \oplus J_1$ both have charpoly $= x^4$ and minpoly $= x^2$, but their invariant factors (x^2, x^2) and (x, x, x^2) differ, so they are not similar.

Both witnesses are machine-verified (the $\varphi \oplus \varphi$ case asserts equal characteristic polynomial, trace, and Mahler measure yet `is_similar` = `False`; commit `c1db881`). This is why the seed-naming bridge reads the Smith normal form rather than the characteristic polynomial.

Remark 3.4 (Exact centroid, with an honest snap fallback). The common case is that the persistent centroid is *itself* an exact algebraic integer of K, so its coordinates are the seed verbatim and Theorem 3.1 returns the exact monic-integer minimal polynomial with no rounding (the worked seeds of this paper are all of this kind). For a genuinely non-integral centroid the implementation falls back to a nearest-integer lattice snap (ties-to-even, computed exactly over $\mathbb{Q}$—no float), clearly labelled `nearest_integer_fallback` in the proposal and re-validated by the same bridge, so a snap that is not an algebraic integer is rejected. The fallback is flagged for a future LLL/PSLQ recovery; either way the column actually adjoined is an exact-Fraction vector whose minimal polynomial is certified by Theorem 3.1, so the exactness of the basis is never compromised. We note the snap here so the framing matches the code, which does not assume the centroid is always exactly integral.

3.2 The disjoint compositum

When the residual lives in a field KL linearly disjoint from K, growth is the substrate's Kronecker construction.

Proposition 3.5 (Kronecker Gram [1, Prop. 4.6])**.** *For linearly disjoint K, L with bases $\{e_i\}, \{f_j\}$, the trace-form Gram of KL on the product basis is $G_{KL} = G_K \otimes G_L$, and $\det G_{KL} = (\det G_K)^{[L:\mathbb{Q}]}(\det G_L)^{[K:\mathbb{Q}]}$.*

Example 3.6 (Adjoining $\sqrt{7}$)**.** $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ ($G_K = \operatorname{diag}(4, 8, 12, 24)$) and $L = \mathbb{Q}(\sqrt{7})$ ($G_L = \operatorname{diag}(2, 14)$) are disjoint, so

$$G_{KL} = G_K \otimes G_L = \operatorname{diag}(8, 56, 16, 112, 24, 168, 48, 336).$$

A $\sqrt{7}$-component is detected by a nonzero field-residual against K (here the score is the out-of-field Gram diagonal 56); growth proposes the disjoint extension $\big($degree $4 \to 8,\ x^2 - 7\big)$ and, on confirm, returns the residual to zero. Machine-verified by `test_p2b_sqrt7_detect_propose_confirm_grows_and_zeros_r` and the Kronecker det-relation test (commits `3e399f1`, `75524b3`).

Remark 3.7 (The determinant check, exactly)*.* The Kronecker construction carries an exact determinant relation, $\det G_{KL} = (\det G_K)^{[L:\mathbb{Q}]}(\det G_L)^{[K:\mathbb{Q}]}$. For Example 3.6, $\det G_K = \det \operatorname{diag}(4, 8, 12, 24) = 9216$ and $\det G_L = \det \operatorname{diag}(2, 14) = 28$, so $\det G_{KL} = 9216^2 \cdot 28^4$, matching the product of the eight Kronecker diagonal entries directly. The identity is a check, not an assumption: it is recomputed exactly by the probe and by the training det relation test (commit `75524b3`), and it is precisely this relation that fails in the non-disjoint case below.

3.3 The non-disjoint compositum (resolving an open problem)

The substrate paper isolated the non-disjoint case and flagged it open: when the candidate generator's minimal polynomial *factors* over K, one has $[K(\beta) : K] < \deg m_\beta$, the Kronecker shortcut is invalid, and "constructing the correct Gram requires a primitive element…and is left open" [1, Rem. 4.7, O4]. We construct it in the bounded-degree case.

Definition 3.8 (Effective degree)**.** For $K = \mathbb{Q}(\alpha)$ of degree m and a candidate algebraic integer β with minimal polynomial m_β over $\mathbb{Q}$, factor m_β over K and let e' be the degree of the irreducible factor that β satisfies. The *effective degree* of the extension is $[K(\beta) : K] = e'$, and the true compositum $K(\beta)$ has degree $m\,e'$ over $\mathbb{Q}$—*not* the tensor degree $m \deg m_\beta$.

Theorem 3.9 (Bounded non-disjoint compositum)**.** *Let $K = \mathbb{Q}(\alpha)$ (degree m) and β an algebraic integer with m_β of degree k. Suppose m_β factors over K as $\prod_\ell g_\ell$ with each $\deg g_\ell \le D_{\max}$ (a declared factorisation bound). Choosing a primitive element $\theta = \alpha + c\beta$ of $K(\beta)$ for a small integer c, the operator minimal polynomial $m_\theta \in \mathbb{Z}[x]$ has degree exactly $m\,e'$, where e' is the degree of the K-factor satisfied by β; the construction selects the correct factor by testing which root reproduces β, and rejects spurious factors arising from the other embeddings of α. The result is an exact monic-integer m_θ and an exact reconstruction of β in the power basis of θ.*

Construction and verification. Factor m_β over K by reducing the problem to factoring the resultant-derived operator polynomial over $\mathbb{Q}$, using a bounded Kronecker factorisation (exact over $\mathbb{Z}$, refusing above `KRONECKER_DEGREE_CAP` $= 12$). The primitive element $\theta = \alpha + c\beta$ generates $K(\beta)$ for all but finitely many c; its minimal polynomial is the operator minimal polynomial of multiplication by θ on $K(\beta)$, degree $[K(\beta) : \mathbb{Q}] = me'$. Among the irreducible factors of the operator polynomial, exactly one has θ as a root with β recoverable as a $\mathbb{Q}$-polynomial in θ; the others come from embeddings of α under which θ collapses to a smaller field and are rejected. The whole computation is exact; it is the machine-checked content of `test_build_compositum_witness`, `test_p2c_witness_grows_true_compositum_not_tensor`, and `test_factor_over_Q_correctness` (commit `3e399f1`). $\square$

Example 3.10 (The degree-4 witness $\mathbb{Q}(\sqrt{2})(\sqrt{2}+\sqrt{3}) = \mathbb{Q}(\sqrt{2},\sqrt{3})$)**.** Take $K = \mathbb{Q}(\sqrt{2}) = \mathbb{Q}[x]/(x^2-2)$ (so $m = 2$) and adjoin $\beta = \sqrt{2}+\sqrt{3}$, whose minimal polynomial over $\mathbb{Q}$ is x^4-10x^2+1 (degree $k = 4$). Over K this *factors*,

$$x^4 - 10x^2 + 1 = (x^2 - 2\sqrt{2}\,x - 1)(x^2 + 2\sqrt{2}\,x - 1) \quad \text{in } \mathbb{Q}(\sqrt{2})[x],$$

so $e' = 2$ and the true compositum has degree $me' = 4$, *not* the tensor degree $2 \cdot 4 = 8$: the Kronecker 8-dimensional Gram would describe a space that does not exist. The construction takes the primitive element $\theta = \alpha + \beta$ ($c = 1$), whose operator minimal polynomial over $\mathbb{Q}$ has degree 6,

$$x^6 - 25x^4 + 91x^2 - 75 = (x^2-3)\,(x^4 - 22x^2 + 25),$$

and selects the genuine factor

$$m_\theta = x^4 - 22x^2 + 25, \tag{4}$$

of degree 4, rejecting the spurious $x^2 - 3$ (whose root $\sqrt{3}$ comes from the embedding $\alpha = -\sqrt{2}$, under which $\theta = -\sqrt{2}+(\sqrt{2}+\sqrt{3}) = \sqrt{3}$ collapses to $\mathbb{Q}(\sqrt{3})$). In the power basis of θ, $\beta = \sqrt{2}+\sqrt{3}$ is recovered exactly as

$$\beta = -\tfrac{7}{20}\,\theta + \tfrac{1}{20}\,\theta^3,$$

i.e. coordinate vector $(0, -\frac{7}{20}, 0, \frac{1}{20})$, and its field-residual against the grown field is 0. All of (4), the degree-6 factorisation, the factor selection, and the exact reconstruction of β are machine-verified (`test_compositum_nondisjoint.py`, commit `3e399f1`).

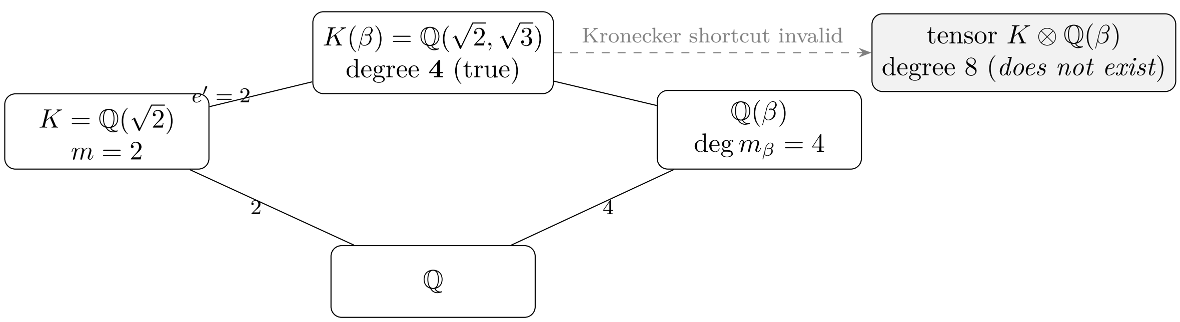

Figure 2: The non-disjoint witness of Example 3.10. Because $m_\beta = x^4 - 10x^2 + 1$ factors over $K = \mathbb{Q}(\sqrt{2})$ (so $e' = [K(\beta) : K] = 2$), the true compositum $\mathbb{Q}(\sqrt{2},\sqrt{3})$ has degree $me' = 4$, not the tensor degree $m \deg m_\beta = 8$; the disjoint Kronecker Gram $G_K \otimes G_{\mathbb{Q}(\beta)}$ would describe an 8-dimensional space that does not exist. The construction selects $m_\theta = x^4 - 22x^2 + 25$ from the degree-6 operator polynomial, machine-verified.

Remark 3.11 (Honest scope of the non-disjoint construction). This resolves the construction in the *bounded-degree* regime: the factorisation over $\mathbb{Q}$ is a Kronecker method capped at degree 12, above which the learner refuses (rather than producing a fictitious Gram) and flags a Zassenhaus/p-adic upgrade as future work. So the open problem O4 of [1] is *advanced and machine-verified in the bounded case*, not settled in full generality. We classify the witness as *exact-computed* (Table 6).

4 Capacity and the applied growth gate

The substrate paper proves the static admissibility theory; here we report the gate as *implemented* and its principled, opt-in tightening.

4.1 The exact Northcott gate

A monic-integer minimal polynomial m is *admissible* for a budget $(D_{\max}, H_{\max})$ iff $\deg m \le D_{\max}$ and $\mathrm{ch}(m) = \max_i |c_i| \le H_{\max}$; the admissible set is finite (Northcott), and the decision is made on *integers only* [1, Prop. 5.5]. The shipped `capacity_decision` returns one of GROW/STOP/REJECT: STOP if the residual norm is at or below a floor (defect captured / noise); else REJECT if inadmissible; else GROW. The float Mahler measure never crosses the boundary—a float coefficient, even an integer-valued one, is refused at the gate.

Example 4.1 (The gate decides on exact integers)**.** The seed $2\sqrt{6}$ (x^2-24) has $(\deg, \mathrm{ch}) = (2, 24)$ and $\sum c_i^2 = 577$, certifying $\mathrm{M}^2 \le 577$ by Landau with no root. Under Budget$(64, 256)$ and gain 96 the decision is GROW; under Budget$(4, 10)$ it is REJECT $(24 > 10)$; with gain 0 it is STOP. The seed $\sqrt{7}$ $(x^2-7, \sum c_i^2 = 50)$ certifies $\mathrm{M} \le 8$ exactly $(50 \le 64)$ while the float Mahler is 7.0000000000...; the integer gate is invariant to that rounding noise. Machine-verified (`test_capacity.py`, commit `c8fdb8a`).

4.2 The Fisher reading and the degree-aware floor

The substrate proves that on the trace-zero (residual) subspace the trace form is degree-scaled Fisher information, $\|\mathbf{r}\|_G^2 = n\,\mathcal{I}_{\exp}(\mathbf{r})$, provided $1 \in \mathrm{col}(B)$ (so $\mathrm{Tr}(\mathbf{r}) = 0$) [1, Lem. 7.9, Thm. 7.11]. The implementation re-verifies the underlying matrices exactly and uses the identity to make a growth floor *degree-aware*.

Example 4.2 (Exact Fisher matrices)**.** With $G = M^{\mathsf{T}}M$ and $\mathcal{I}_{\exp} = \frac{1}{n}\big(G - \frac{1}{n}tt^{\mathsf{T}}\big)$ (t the trace vector):

$$\begin{array}{lll} \mathbb{Q}(\sqrt{5}),\ \{1,\varphi\}: & G = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right), & \mathcal{I}_{\exp} = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 5/4 \end{smallmatrix}\right); \\ \mathbb{Q}(\sqrt{2},\sqrt{3}): & G = \mathrm{diag}(4, 8, 12, 24), & \mathcal{I}_{\exp} = \mathrm{diag}(0, 2, 3, 6); \\ \mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{7}): & G = \mathrm{diag}(8, 56, 16, 112, 24, 168, 48, 336), & \mathcal{I}_{\exp} = \mathrm{diag}(0, 7, 2, 14, 3, 21, 6, 42). \end{array}$$

For the residual $\sqrt{5} = 2\varphi - 1 = (-1, 2)$ (trace-zero), $\|\sqrt{5}\|_G^2 = 10 = 2 \cdot 5 = n\,\mathcal{I}(\sqrt{5})$ with $n = 2$. On every trace-zero direction $G = n\,\mathcal{I}_{\exp}$—a *subspace* identity, not a full-matrix one: $\mathcal{I}_{\exp}$ and $\frac{1}{n}G$ agree on the trace-zero directions but differ at the constant ($\mathcal{I}_{\exp}$ vanishes there, while $\frac{1}{n}G$ does not), which is exactly why the identity requires $1 \in \mathrm{col}(B)$. Machine-verified, both directions (`test_a3p2_fisher.py`, commit `d526098`).

Remark 4.3 (The applied threshold, with an honest reciprocity caveat). The opt-in information threshold (default *off*; with it off the gate returns the shipped verdict byte-for-byte and imports no interval-arithmetic library) tests gain $\|\mathbf{r}\|_G^2$ against cost $\lambda \log \mathrm{M}(\theta)$ and a floor. The floor branch keys on whether the seed is *reciprocal*:

- *Non-reciprocal* seed: the floor is Smyth's $\mu_S = 1.32471795724\ldots$ (real root of $x^3 - x - 1$), a *theorem* [2]; stored as a certified rational bracket. With $\lambda = 2$ the constant floor is $2\log\mu_S = 0.5623991486\ldots$, the degree-aware floor $n \cdot 2\log\mu_S$ is 2.2496 at $n = 4$ and 4.4992 at $n = 8$.
- *Reciprocal* seed (Salem numbers, units—which arise naturally): there is *no proven uniform positive floor* (the unsolved core of Lehmer's problem); the gate falls back to Dobrowolski's degree-dependent bound [4], which is vacuous at small degree. This caveat is explicit in the code and the tests, and is recorded as *open* in Table 6.

The threshold *subsumes* the shipped growth cases (gain $96, 56 \gg$ cost), and the degree-aware floor is what differs from the floor-0 shipped gate on lattice-aligned sub-threshold noise. Machine-verified (`test_capacity_threshold.py`, `test_a3p2b_threshold.py`, commits `c8fdb8a`, `c5605d7`); the exchange rate is no longer a posited scalar but the derived identity $\lambda = 2c$ (Theorem 4.6), with c the conformal constant of the declared generative model. The shipped $\lambda = 2$ is its Model–1 ($c = 1$, Gaussian) instance; the degree–aware floor $n \cdot 2 \log \mu_S$ above is its Model–2 ($c = n$) instance, so the two floor families of this remark are the two conformal readings of one identity (Remark 4.7).

Example 4.4 (A certified GROW, no trust in a rounded value)**.** The information half of the threshold is intrinsically analytic ($\log \mathrm{M}$ is transcendental), but a *decision* can still be certified: an exact rational gain need only dominate a rigorous *upper* bound on the cost. For $\sqrt{7}$ ($x^2 - 7$, $\mathrm{M} = 7$) at $\lambda = 2$, interval arithmetic gives $\log 7 \in [1.94591, 1.94592]$, so $\lambda \log 7 \le 2 \cdot 1.94592 = 3.89184$, and the exact gain $\|\mathbf{r}\|_G^2 = 56$ satisfies $56 \ge 3.89184 \ge \lambda \log 7$: GROW is certified with no trust in any rounded value. The same procedure certifies $2\sqrt{6}$ ($x^2 - 24$): $\log 24 \in [3.17805, 3.17806]$ gives $\lambda \log 24 \le 6.35612 \le 96$. By contrast, lattice-aligned noise of exact magnitude $\frac{1}{10}$ is STOPped by the Smyth constant floor, $\frac{1}{10} < 0.5624$. The enclosures are produced by *verified interval arithmetic* (`mpmath.iv`, the same discipline as the implementation's gate), not by a trusted rounded log: the companion probe asserts the exact rational gain dominates the rigorous interval *upper* bound (`test_certified_grow_enclosures`, §A), and the production threshold is exercised by `test_capacity_threshold.py` (commit `c8fdb8a`).

Remark 4.5 (Exactness has a price, but never changes a decision). Every gate quantity above is exact integer or rational arithmetic, trading floating speed for reproducibility. The standing practical caveat is rational coefficient growth: exact Gram and projector entries can acquire large numerators and denominators, so an implementation bounds the working degree and height (which the capacity gate already enforces) or tests equalities modulo a prime and lifts. The only inexact step anywhere is the threshold comparison against $\log \mathrm{M}$, handled by the certified enclosure of Example 4.4 while the gain side stays an exact rational. None of this alters a decision—it only prices exactness.

4.3 The exchange rate, derived: $\lambda = 2c$

The threshold of Remark 4.3 compares a *gain* $\|\mathbf{r}\|_G^2$ against a *cost* $\lambda \log \mathrm{M}(\theta)$. We now show the rate λ is not a free knob: reading the trace form as a Fisher metric turns "grow iff gain $\ge$ cost" into a two–part minimum–description–length (MDL) code [17], and the rate falls out as an identity. This closes open problem (O1) of §10 and, together with Čencov's theorem, recasts (O4) precisely.

Theorem 4.6 (The exchange rate is twice the conformal constant)**.** *Read the trace form as the Fisher metric $\mathcal{I} = G/c$ of the declared generative model—$c = 1$ for the Gaussian location family $\mathcal{N}(Ma, I_n)$ and $c = n$ for the maximum–entropy family on the Galois conjugates restricted to the trace–zero subspace, the two readings established in [1]. For a residual $\mathbf{r}$, the second–order expansion of the Kullback–Leibler divergence between the captured model and the model that also explains $\mathbf{r}$ is*

$$D_{\mathrm{KL}} = \tfrac{1}{2}\,\mathbf{r}^{\mathsf{T}} \mathcal{I}\,\mathbf{r} = \frac{1}{2c}\,\|\mathbf{r}\|_G^2. \tag{5}$$

The two–part MDL balance "adjoin a direction iff the bits it explains meet the bits it costs," $D_{\mathrm{KL}} \ge \log \mathrm{M}(\theta)$, is therefore exactly *the threshold $\|\mathbf{r}\|_G^2 \ge \lambda \log \mathrm{M}(\theta)$ of Remark 4.3, with*

$$\boxed{\lambda = 2c}\,. \tag{6}$$

The "2" is the structural $\frac{1}{2}$ of the quadratic KL expansion, inverted; it is not a parameter. The entire residual freedom is the conformal constant c.

Proof. The Fisher information is the Hessian of D_{KL} at the matching point, so the second–order term of the expansion in a parameter perturbation δ is $\frac{1}{2}\delta^{\mathsf{T}}\mathcal{I}\,\delta$: the zeroth–order term vanishes, and the first–order term vanishes because the score has mean zero, so the leading nonzero term is quadratic. Setting $\delta = \mathbf{r}$ and $\mathcal{I} = G/c$ gives (5) (for the Gaussian location family the expansion is *exact*, KL being already quadratic). Substituting into $D_{\mathrm{KL}} \geq \log \mathrm{M}$ and clearing the factor $\frac{1}{2c}$ yields $\|\mathbf{r}\|_G^2 \geq 2c\log\mathrm{M}$, i.e. $\lambda = 2c$. The per–seed entropy $\log\mathrm{M}$ is the description length of the adjoined direction (the topological entropy of the companion toral automorphism, Lind–Schmidt–Ward [15]); it is the *cost* ledger and does not enter λ, which is purely the unit conversion between the gain $\|\mathbf{r}\|_G^2$ and the explained divergence D_{KL}. □

Remark 4.7 (The two shipped floors are the two readings of $\lambda = 2c$)*.* Remark 4.3 already carries both instances. The *constant* floor $2\log\mu_S = 0.5623991\ldots$ is the Model–1 value $\lambda = 2$ ($c = 1$); the *degree–aware* floors $n \cdot 2\log\mu_S$ (2.2496 at $n = 4$, 4.4992 at $n = 8$) are the Model–2 value $\lambda = 2n$ ($c = n$). The single expression $2c\log\mu_S$ evaluated at $c \in \{1, n\}$ reproduces both, so the paper's two floor families were never independent choices: they are the two conformal readings of one identity. (Verified: the floors recompute to $0.5624, 2.2496, 4.4992$ from μ_S, the root of $x^3 - x - 1$, in the companion derivation check of §A.)

Remark 4.8 (The residual freedom is the conformal scale, and it is irreducible)*.* Theorem 4.6 replaces the posited scalar λ by the identity $\lambda = 2c$, leaving a *single* free quantity: the conformal constant c. By Čencov's theorem the Fisher metric is the unique Riemannian metric invariant under sufficient statistics only *up to a positive scale* [16]; that scale is precisely $c = \lambda/2$. Hence no invariance principle internal to information geometry can fix c: this is an impossibility result, not a missing lemma, and it is exactly the content of the no–canonical–model conjecture of [1]. Two readings of the field's own structure already name two values: $c = 1$, under which the Jeffreys information volume equals $\sqrt{\det G} = \sqrt{|d_K|}$ (the field discriminant) [18, 1], and $c = n$, under which per–conjugate significance is degree–invariant. Choosing c is choosing a generative model; it is a declared modeling commitment. Definition 4.10 adds a third value, native not to an external prior but to the geometry of a critical point.

We make the critical–point reading precise through the self–action of the C–ladder, the quadratic family $x^2 + x = C$ whose discriminant $1 + 4C$ controls the keystone (the golden case $C = 1$ gives discriminant 5).

Proposition 4.9 (Self–action spectrum of the C–ladder)**.** *For $C \in \mathbb{Q}$ let $R_C = \left(\begin{smallmatrix} 0 & C \\ 1 & -1 \end{smallmatrix}\right)$ be the companion matrix of $x^2 + x - C$, with eigenvalues $(-1 \pm \sqrt{1+4C})/2$. The Lie self–action $X \mapsto [R_C, X] = R_C X - X R_C$ on $M_2(\mathbb{Q})$ has spectrum*

$$\left\{ -\sqrt{1+4C},\ 0,\ +\sqrt{1+4C} \right\} \qquad (\text{the } 0 \text{ with multiplicity } 2), \tag{7}$$

its nonzero eigenvalues being the difference of the two eigenvalues of R_C. The 0–eigenspace is the centraliser $\operatorname{span}\{I, R_C\}$*. At the golden value $C = 1$ the spectrum is $\{-\sqrt{5}, 0, +\sqrt{5}\}$ with $\sqrt{5} = \varphi - \psi$, and R_1 is conjugate to the keystone $R = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ (*$\mathrm{charpoly} = x^2 - x - 1$*; both have discriminant* 5*, hence the same self–action gap).*

Proof. For a 2×2 matrix with eigenvalues $\{a, b\}$ the adjoint action has eigenvalues $\{a - a, a - b, b - a, b - b\} = \{0, 0, \pm(a - b)\}$; here $a - b = \sqrt{1+4C}$. The centraliser of a non–scalar 2×2 matrix is the two–dimensional algebra $\operatorname{span}\{I, R_C\}$, which is the kernel of the action. The golden specialisation is $1 + 4 \cdot 1 = 5$, and $\varphi - \psi = \frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2} = \sqrt{5}$. □

The three eigen–channels of (7) are exactly the growth decision read in the eigenframe: $+\sqrt{1+4C}$ is the off–diagonal *growth* channel (GROW), $-\sqrt{1+4C}$ the decay channel (STOP), and the marginal 0–channel $\operatorname{span}\{I, R_C\}$ is the captured, residual–zero direction. This is the structure that fixes the conformal scale at a critical point.

Definition 4.10 (The critical–point (frame–shift) canonicalization)**.** At a critical value C of the ladder, read the growth decision in the eigenframe of the self–action $[R_C, \cdot]$. Its intrinsic scale is the spectral gap $\sqrt{1+4C}$ of Proposition 4.9—the distance from the captured (0) channel to the growth/decay channels, and also $\sqrt{\mathrm{M}(\sqrt{1+4C})}$ since $\mathrm{M}(x^2 - (1+4C)) = 1+4C$. Equating the decision scale λC with this gap,

$$\lambda\, C = \sqrt{1+4C} \quad\Longrightarrow\quad \lambda = \frac{\sqrt{1+4C}}{C}, \qquad c = \frac{\lambda}{2} = \frac{\sqrt{1+4C}}{2C}, \tag{8}$$

fixes the conformal constant at the gate. At the golden gate $C = 1$ this is $c = \sqrt{5}/2$, $\lambda = \sqrt{5}$—the exchange rate *is* the spectral gap.

Remark 4.11 (What is closed, and what is a declared choice)*.* The identity $\lambda = 2c$ is *forced* (Theorem 4.6): the exchange rate is no longer free, and (O1) and (O4) of §10 are one degree of freedom, related by $\lambda = 2c$. What remains is not a *value* but a *model*: which c in Table 2 the application declares. Čencov's theorem guarantees this cannot be reduced further by invariance. The critical–point value $\sqrt{1+4C}/(2C)$ is the one the substrate's own structure supplies—the eigenframe's intrinsic scale rather than an imported prior—but it is, per Čencov, a principled *choice*, not a derivation. Operationally the choice is immaterial on the shipped seeds: their gains $96, 56$ exceed every floor for all $c \in [1, n]$, so the shipped GROW/STOP verdicts are invariant across the named models (it would matter only for a seed whose gain lands in the band $(2\log\mu_S,\ 2n\log\mu_S)$).

Remark 4.12 (The discriminant flip and the totally–real hypothesis)*.* The gap $\sqrt{1+4C}$ is real precisely when $1+4C > 0$. The discriminant $D = 1+4C$ crosses zero at $C = -\frac{1}{4}$ (a double root), and for $D < 0$ the eigenvalues of R_C become complex conjugates on a circle: the $\pm$ channels of (7) turn from real growth/decay into imaginary rotation $\pm i\sqrt{|D|}$, while the captured 0–channel persists. This is the same sign that governs the metric: the trace form of $\mathbb{Q}(\sqrt{D})$ is positive definite (a Fisher metric) for $D > 0$ (totally real) and indefinite for $D < 0$ (a complex place), which is why the Fisher reading of §4.3 and [1] requires the totally–real hypothesis. The conformal canonicalization of Definition 4.10 therefore lives on the $D > 0$ side; the $D < 0$ regime is a rotation, not a growth, and is recorded as out of scope here.

5 The automatic learning loop

The pieces above are driven automatically by a calibration-gated detector that never auto-acts, and a field-growing learner that measures its own gain.

5.1 A calibration-gated detector

The detector scores each exact observation by the residual norm $s = \|\mathbf{r}\|_G^2$ and raises an *anomaly* only when $s >$ threshold *and* a calibration predicate passes; it then surfaces a growth *proposal* (a `SeedProposal` in-field, a `FieldExtensionProposal` out-of-field), never auto-confirming. Formally `anomaly` $=$ `novel` $\wedge$ `calibrated`, and the wrapped models' basis size and field degree are invariant until an explicit caller confirm.

Proposition 5.1 (Propose, never auto-grow)**.** *For an in-distribution stream (every observation in* $\mathrm{col}(B)$*) the score is* 0 *and nothing is surfaced. For a persistent off-axis novelty the score is positive; while uncalibrated the novelty is recorded but the alert is* suppressed *and no proposal surfaces; once calibrated the same observation alerts and surfaces a proposal. Scoring and surfacing leave the wrapped models unchanged. Surfaced proposals are appended to a* SHA-256 *witness chain* (3)*.*

Proof. Each clause is a machine-checked assertion: `test_a7_in_distribution_stream_no_anomaly` (score 0, nothing surfaced), `test_a7_calibration_gate_suppresses_until_calibrated`, `test_a7_proposes_bu` and `test_a7_witness_of_surfaced_proposals` (verify `True`; tamper a recorded score and verify flips `False`); commits `6ad20eb`, `038ee27`. □

Remark 5.2 (What "calibration" is, precisely—and what it is not)*.* The calibration predicate is an *injected* gate, and the shipped constructor default is *permissive* (it admits once persistence is met). So "calibration-gated" names the *mechanism*—detection is decoupled from alerting through an explicit gate—not a substantive default policy; we say so plainly rather than imply a calibration the default does not perform. A concrete, exact instance is provided and tested, `variance_calibration`, which opens only once the off-axis accumulator has both warmed up *and settled*: its accumulated per-coordinate Welford variance $\sum_i M2_i$ (previously computed but unused) lies within a declared fraction τ of the centroid's squared magnitude,

$$\textstyle\sum_i M2_i \ \le\ \tau\, n \sum_i \mathrm{mean}_i^2, \tag{9}$$

all exact Fraction. This is distinct from persistence: persistence tests that the residual *direction* holds (the alignment test of Algorithm 1), calibration that its *magnitude* has settled, so an aligned-but-volatile stream persists yet stays uncalibrated and surfaces no proposal. Machine-verified by `test_calibration.py` (a stable stream calibrates after warm-up; a volatile 1×/10× stream persists but is blocked from proposing; commit `1366d1c`).

Example 5.3 (A streaming detector trace)**.** In the ambient $K = \mathbb{Q}(\sqrt{2}+\sqrt{3})$ with forced basis $B = \{1, \theta\}$ and a calibration predicate that opens after the third observation, the off-axis $2\sqrt{6} = \theta^2 - 5$ carries the exact score 96, and an in-span observation scores 0:

t	observation	$\lVert\mathbf{r}\rVert_G^2$	calibrated	action
1	$2-\theta$ (in $\mathrm{col}\,B$)	0	no	—
2	$3+4\theta$ (in $\mathrm{col}\,B$)	0	no	—
3	$1+2\sqrt{6}$ (off-axis)	96	no	novelty recorded, alert *suppressed*
4	$1+2\sqrt{6}$	96	yes	alert; *propose* seed x^2-24
5	confirm	—	yes	basis grows to $\{1, \theta, 2\sqrt{6}\}$; witnessed
6	$1+2\sqrt{6}$	0	yes	captured

The score is an exact rational at every step; the basis is mutated only at $t = 5$, and only by an explicit *confirm*; steps 3–4 show calibration decoupling detection from alerting, and step 6 shows capture collapsing the score to 0. Machine-verified by the detector tests (commit `6ad20eb`).

5.2 Detector-driven gain and the full loop

The seam that makes the loop automatic is that the growth *gain* is not supplied by the caller but *measured*: the field-growing learner computes the exact out-of-field residual and that value *is* the gain.

Theorem 5.4 (The automatic loop returns the residual to zero)**.** *Starting from $K = \mathbb{Q}(\sqrt{2})$ with $B = \{1\}$ (so $1 \in \mathrm{col}(B)$, degree 2), the loop*

$$\textit{detect} \to \textit{auto-gain} \to \textit{propose} \to \textit{gate(effective degree)} \to \textit{confirm} \to \textit{re-home onto } \mathbb{Q}(\theta) \to \mathbf{r} = \mathbf{0}$$

captures an out-of-field $\beta = \sqrt{2} + \sqrt{3}$: the auto-gain equals the measured residual, the gate judges the true *compositum degree 4 (Definition 3.8), and after confirm the basis is re-homed into the power basis of the primitive θ with $m_\theta = x^4 - 22x^2 + 25$, returning the residual to 0 exactly.* confirm *is the sole field-growth mutator;* propose *leaves the degree unchanged.*

Proof. The auto-gain is the exact field-residual: an exact `Fraction` (value 12 for the witness), and scaling the observed element by 2 scales it quadratically to 48, confirming it is a genuine norm rather than a tag. The gate uses `effective_degree` $= me' = 4$, so $\mathrm{Budget}(D_{\max} = 4)$ grows while $\mathrm{Budget}(D_{\max} = 3)$ refuses with `capacity_reject`. Re-homing recomputes each forced column in the power basis of θ (Theorem 3.9) and clears denominators (Lemma 5.5); the captured directions then have residual 0. Machine-verified end-to-end by `test_e2e_out_of_field_residual_driven_to_zero_throu` `test_seam_detector_residual_is_the_growth_gain`, `test_auto_gain_equals_measured_residual`, and `test_effective_degree_is_true_compositum_degree` (commits `038ee27`, `ef3b86d`). □

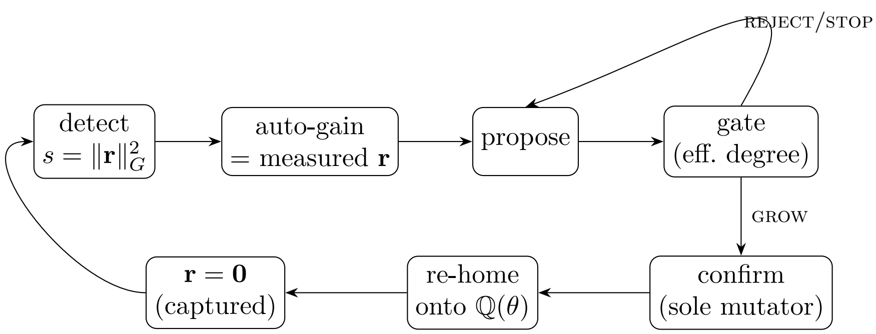

Figure 3: The automatic loop of Theorem 5.4. The detector's measured residual *is* the growth gain; the gate judges the *true* compositum degree (Definition 3.8); *confirm* is the sole mutator; re-homing onto the primitive element's power basis returns the residual to 0. The gate may instead REJECT (inadmissible) or STOP (sub-floor), leaving the model unchanged.

Lemma 5.5 (Re-home is projector-invariant under column scaling)**.** *Re-homing a forced-basis column from K into $\mathbb{Q}(\theta)$ can produce fractional power-basis coordinates; scaling a column to the smallest integer vector on the same line (clearing denominators) does not change* $\mathrm{col}(B)$*, hence does not change the projector P or any residual. In particular a captured direction stays captured across the re-home.*

Proof. $P = B(B^\top G B)^{-1} B^\top G$ depends only on $\mathrm{col}(B)$: replacing a column b by cb ($c \in \mathbb{Q}^\times$) is the right-multiplication $B \mapsto BD$ by an invertible diagonal D, under which P is invariant (P is the unique G-orthogonal projector onto $\mathrm{col}(B)$). Machine-checked by `test_rehome_preserves_subspace_with_a_nor` (commit `ef3b86d`). □

Remark 5.6 ($1 \in \mathrm{col}(B)$ survives growth)*.* The constant $1 = (1, 0, \ldots, 0)$ is the zeroth power-basis vector in *any* power basis, so $1 \in \mathrm{col}(B)$ is preserved across the re-home into $\mathbb{Q}(\theta)$, keeping the Fisher precondition of §4 valid in the grown field. Machine-checked by `test_one_in_basis_preserved_across_growth` (commit `ef3b86d`). On this increment the info-threshold is left off (floor 0), so a tiny positive gain still grows; the principled floor is the opt-in of Remark 4.3.

6 A residual-valued language

We now exhibit the same residual-return mechanism as a *language*, over the Clifford algebra $\mathrm{Cl}(2,0) \cong M_2(\mathbb{R})$, realised exactly over $\mathbb{Q}$ (`Fraction`, no float in any decision). This layer lives in a separate, one-way module; it reads the number-field engine only through a φ-keystone bridge and is never imported back.

6.1 The exact carrier

A *holding* is $X = a \cdot 1 + b\,e_1 + c\,e_2 + d\,i$ with $a, b, c, d \in \mathbb{Q}$, $e_1^2 = e_2^2 = 1$, $i = e_1 e_2$, $i^2 = -1$, and the matrix isomorphism $\mathrm{mat}(X) = \left(\begin{smallmatrix} a+c & b-d \\ b+d & a-c \end{smallmatrix}\right)$. The readings $\nu(X) = M(X) - X$ (idempotence defect, $M(X) = \tilde{X}X$ the Gram) and $R_K(X) = M(X) - \tau(M(X))\,1$ (trace-free Gram) are decided at zero tolerance: $\nu(P_0) = \mathbf{0}$ for the gate $P_0 = \frac{1}{2}(1 + e_2)$, $R_K(1) = \mathbf{0}$, $R_K(P_0) = \frac{1}{2}e_2 \neq \mathbf{0}$, and $R_K^2 = \mathbf{0}$ (bite-depth ≤ 2). Cayley–Hamilton $\Phi_X(X) = X^2 - \tau(X)X + \det(X)\,1 = \mathbf{0}$ holds exactly. Machine-verified (`test_holding.py`, commit `f925e50`; exact core `6cd3800`).

Example 6.1 (The readings are zero-tolerance decisions)**.** The gate $P_0 = \frac{1}{2}(1 + e_2)$ is a symmetric idempotent: $\nu(P_0) = M(P_0) - P_0 = \mathbf{0}$ exactly, so P_0 is "captured" by the idempotence reading, yet $R_K(P_0) = \frac{1}{2}e_2 \neq \mathbf{0}$—it is not conformal, so the trace-free Gram is a nonzero residual. By contrast $R_K(1) = \mathbf{0}$ (the unit is conformal). The type lattice is exact: the keystone R has $\tau(R) = \frac{1}{2}$, $\det(R) = -1$, hence $\mathrm{TYPE}(R) = \{\mathrm{gen}\}$ (it satisfies the golden law $X^2 = X + 1$), while P_0 has $\mathrm{TYPE}(P_0) = \{\mathrm{idem}, \mathrm{rest}\}$ and a generic holding has empty type. Every one of these is an == decision over $\mathbb{Q}$, machine-verified (`test_holding.py`, `test_type_lattice_exact`, commits `f925e50`, `4817b62`).

6.2 The φ-keystone return operator

The keystone is $R = \frac{1}{2} + e_1 - \frac{1}{2}e_2$, i.e. $\mathrm{mat}(R) = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, the companion matrix of $x^2 - x - 1$. It satisfies the golden law $R^2 = R + 1$ (so $\mathrm{TYPE}(R) = \{\mathrm{gen}\}$), and $\mathrm{mat}(R)$ is symmetric. The *return operator* is the Sylvester self-action

$$\mathcal{L}(X) = RX + XR - X. \tag{10}$$

Proposition 6.2 ($\mathcal{L}$ is exact and oracle-checked)**.** *$\mathcal{L}$ is computed by the exact geometric product and agrees on every sampled holding with an* independent *matrix-route oracle (own* mat*/multiply/*cl*, never calling the holding product), ruling out a single-route bug. $R \notin \ker \mathcal{L}$ (indeed $\mathcal{L}(R) = \frac{5}{2} + e_1 - \frac{1}{2}e_2 \neq \mathbf{0}$).*

Proof. Machine-checked by `test_L_matches_matrix_oracle_exact` and `test_keystone_R_not_in_kernel_exac` (commit `4817b62`). □

Theorem 6.3 (The exact kernel is the lexicon space)**.** *$\ker \mathcal{L}$ is exactly two-dimensional, computed by hand-rolled rational Gaussian elimination on the 4×4 matrix of $\mathcal{L}$ over $\{1, e_1, e_2, i\}$ (not by floating SVD), with exact basis*

$$\ker \mathcal{L} = \mathrm{span}\{\, e_1 + 2e_2, \quad i \,\} = \mathrm{span}\{H(0,1,2,0),\ H(0,0,0,1)\}.$$

The Sylvester eigenvalue argument confirms the dimension: $\mathcal{L} = S - \mathrm{id}$ *where* $S(X) = RX + XR$ *has eigenvalues* $\lambda_a + \lambda_b$ *over the eigenvalues* $\{\varphi, \psi\}$ *of* R *(with* $\varphi + \psi = 1$*), i.e.* $\{2\varphi, 1, 1, 2\psi\}$*, so* $\mathcal{L}$ *has eigenvalue* 0 *with multiplicity* 2.

Proof. The rational nullspace computation returns the stated integer-normalised basis; the eigenvalue count matches. Machine-verified by `test_kernel_exact_dim_and_basis` (commit `4817b62`). $\square$

Remark 6.4 (A corrected hint, pinned by its disproof). An earlier design note claimed the antisymmetric generator "$N = H(0, -1, 1, 0)$" lay in $\ker \mathcal{L}$. This is *false*: $\mathcal{L}(H(0, -1, 1, 0)) = H(-3, 0, 0, 0) \neq \mathbf{0}$. The slip transcribed the four *matrix* entries of $N = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ directly into holding coordinates instead of through the iso; the correct image is $\mathrm{cl}\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = i = H(0, 0, 0, 1)$, which *is* in the kernel. The disproof is itself a machine-checked test (`test_corrected_handoff_hint_is_not_in_kernel`, commit `4817b62`)—a small instance of the discipline that load-bearing claims, including negative ones, are pinned by computation.

6.3 The commit, the word, and the growing lexicon

The *commit* is the exact idempotent orthogonal projector onto $\ker \mathcal{L}$, value-pinned (idempotence alone is also satisfied by the identity, so the matrix is asserted, not merely $P^2 = P$):

$$\mathrm{commit} = \mathrm{proj}_{\ker \mathcal{L}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{1}{5} & \frac{2}{5} & 0 \\ 0 & \frac{2}{5} & \frac{4}{5} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{(in coordinates } 1, e_1, e_2, i\text{)}. \tag{11}$$

A token's *word* is a sign-aware sparse code over the kernel basis (an exact, sqrt-free L^1-relative threshold $\tau = \frac{1}{4}$); sign discriminates antipodal rays, so $\mathrm{word}(i) = (+i)$ and $\mathrm{word}(-i) = (-i)$ are opposite meanings.

Theorem 6.5 (Generalisation by return-to-zero)**.** *A learned dictionary value is exactly a committed residue:* $\mathrm{commit}(X) \in \ker \mathcal{L}$*, decided over* $\mathbb{Q}$*. Tokens that commit to the* same *exact residue share one lexicon entry; tokens with* distinct *residues stay distinct. Thus* E_1 *and* $E_1 + 1$ *merge (the constant* 1 *is orthogonal to the slack, so* $\mathrm{commit}(E_1 + 1) = \mathrm{commit}(E_1) = H(0, \frac{1}{5}, \frac{2}{5}, 0)$*), and positive scaling preserves the word (*$\mathrm{word}(3E_1) = \mathrm{word}(E_1)$*); but* i *(value* $H(0, 0, 0, 1)$*) and* $2i$ *(value* $H(0, 0, 0, 2)$*), though they share the word* $(+i)$*, are* distinct *entries—dedup is on the exact value, not the coarse sign-word. Lexicon identifiers are deterministic content hashes* id $=$ SHA-256(*canonical* Fraction *string*)[0:16]*, order- and run-independent.*

Proof. Machine-verified by `test_projector_exact_matrix_value`, `test_generalization_same_residue_same_` `test_distinct_residues_stay_distinct`, and `test_ids_are_deterministic_and_order_independent` (commits `4817b62`, `d1312ed`). $\square$

Example 6.6 (A growing lexicon, traced)**.** Feeding five tokens through *acquire* (commit, then add) grows the lexicon to four entries: the constant 1 is orthogonal to the slack, so E_1 and $E_1 + 1$ return to the same value and merge; the pseudoscalar i, its double $2i$, and its antipode $-i$ are three distinct exact values—even though i and $2i$ share the word $(+i)$.

token X	committed value $\in \ker \mathcal{L}$	word	lexicon
$E_1 = (0, 1, 0, 0)$	$H(0, \frac{1}{5}, \frac{2}{5}, 0)$	$(+e_1{+}2e_2)$	new entry #1
$E_1 + 1 = (1, 1, 0, 0)$	$H(0, \frac{1}{5}, \frac{2}{5}, 0)$	$(+e_1{+}2e_2)$	merged (same residue)
$i = (0, 0, 0, 1)$	$H(0, 0, 0, 1)$	$(+i)$	new entry #2
$2i = (0, 0, 0, 2)$	$H(0, 0, 0, 2)$	$(+i)$	new entry #3 (same word, distinct value)
$-i = (0, 0, 0, -1)$	$H(0, 0, 0, -1)$	$(-i)$	new entry #4 (antipode)

Five tokens, four entries: generalisation is collapse to a shared *exact* residue, and the sign-word is a label, not the dedup key. Machine-verified by the lexicon tests (commit `d1312ed`).

6.4 The firewall, persistence, and the shell

Statements carry a jurisdiction (Table 3), and only THEOREM/COMPUTED cross a wire as fact; a learned lexicon entry is COMPUTED, never promoted to THEOREM.

The lexicon persists through a SHA-256 hash chain (3) identical in form to the learner's, with exact round-trip and tamper-evidence; and a JSON-in/JSON-out shell exposes the surface (`read`/`render`/`observe`/`propose`/`commit`/`persist`/`restore`) under local-equivalence (in-process = subprocess) and a robustness guard (a malformed request soft-errors, never crashes). The whole layer is one-way disjoint: a scan finds the number-field repository references the language module *nowhere*, and only the keystone bridge reaches the live engine. The full suite is 121 **tests green** at the readiness tag `b1-ready` (commit `a5892c7`); the readiness gate itself is mutation-proven load-bearing.

7 The unifying principle and the φ keystone

7.1 One mechanism, two instances

The two faces are the same five-part structure (Table 4). The deepest shared piece is not analogy but a literal identity: the witness chain (3) is byte-identical in construction across the learner and the lexicon, and capture is exact vanishing of a residual in both.

7.2 The keystone identity

Proposition 7.1 (The φ-keystone is one object)**.** *The companion matrix* $\rho(\varphi) = C(x^2 - x - 1) = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ *of the substrate's first field* $\mathbb{Q}(\varphi)$*, the symmetric matrix* R *of the language return operator* (10) *(with* $R^2 = R + I$*, atom of the golden law), and the keystone Clifford element* $\mathrm{cl}(0.5, 1, -0.5, 0)$ *are the* same *object, the void law* $x^2 = x + 1$*. The three coincide and are verified equal three independent ways: the exact integer companion route (loom), the Clifford route (*$R^2 = R + 1$ *exactly), and the live Mahler measure* $\mathrm{M} = \varphi$*.*

Proof. The bridge module confirms against the live engine: companion $= \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, minimal polynomial $[1, -1, -1] = x^2 - x - 1$, Clifford coordinates $(\frac{1}{2}, 1, -\frac{1}{2}, 0)$, and Mahler measure within tolerance of φ. Machine-verified by the three-route closure test on the keystone and the bridge agreement test (commits `371fe09`, `a5892c7`); the keystone Clifford element satisfies $\mathrm{TYPE}(R) = \{\mathrm{gen}\}$ exactly (commit `4817b62`). ☐

Remark 7.2 (Why φ, and why it matters)*.* The golden field $\mathbb{Q}(\varphi)$ is the substrate's running example (its Gram $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, discriminant 5) and Smyth's floor is the root of $x^3 - x - 1$, a sibling of $x^2 - x - 1$

in the same height theory; the language's entire return dynamics are built on the companion of $x^2 - x - 1$. The keystone is thus not decorative: it is the single arithmetic object through which the learning geometry and the language algebra are the *same* construction seen twice. This is the precise content of Figure 1.

7.3 Persistence is exact re-derivation, on both faces

A consequence of exactness is that a trained model is, in its entirety, a short replayable record rather than an opaque parameter blob, and this holds identically on both faces. On the learning face the state is a list of monic-integer seed minimal polynomials (the forced anchors) plus the residual deltas since the last anchor; because G and P are recomputed bit-for-bit from the anchors (Theorem 3.1), a restart is *re-derivation*, not deserialisation, so a model and its restart agree as operators, not merely to within a tolerance. On the language face the lexicon persists through the same SHA-256 hash chain (3) and restores by re-deriving each entry through the acquisition loop and cross-checking its id, coordinates, and word against the stored record—an exact round-trip that a tampered store fails. In both cases the witness chain makes the whole history replayable and tamper-evident, since each growth step is reproducible from its seed (or token) by exact arithmetic. This is the precise sense in which residual return is exact "all the way down": the persisted object is arithmetic that regenerates itself.

8 Proven computation: every claim, its test, its hash

This is the paper's orientation made explicit. The exactness methodology is uniform across both faces, and every load-bearing claim is pinned to a named test and a commit hash.

8.1 The methodology

(M1) **Exact-rational discipline (no float in a decision).** Every membership, growth, dedup, and kernel decision is taken over $\mathbb{Q}/\mathbb{Z}$ with `Fraction`; a `float` input is refused at intake (a `TypeError`); floats appear only in display fields. Verified by float-rejection tests on both faces.

(M2) **Independent oracles.** Load-bearing operators are cross-checked by a second, independent route: the regular representation against a multiplication-table build, the return operator against a matrix-route oracle, the witness digest against a from-scratch recomputation.

(M3) **Mutation-tested, load-bearing tests.** Tests pin exact *values*, not loose bounds (e.g. the projector matrix (11), not merely $P^2 = P$), and the readiness gate was mutation-proven: planted firewall, one-way, and exact regressions each trip a test.

(M4) **Build / independent-verify / greenlight.** Each increment followed a fixed protocol: *claim* the next exact step, *build* it test-first, *gate* it (the increment's tests plus the full suite plus, for the substrate side, the zero-free-parameter gate), *report* the design and diff, and *hold* for an independent re-derivation before committing. The independent verifier re-derived the load-bearing objects from scratch—the trace-form Gram and projector, the invariant-factor bridge, the return-operator kernel, the hash-chain digest—by a second route, and only an explicit greenlight released the commit. The coordination journal and the commit history are the audit trail; the language layer alone carried this loop through seven gated increments to the `b1-ready` tag.

8.2 What "load-bearing" means: three tests, verbatim

A pointer table is only as good as the tests it points to. Here are three of them in full, reproduced from the companion probe (§A), so the reader can see that they pin *exact values*, not loose bounds. First, the language commit projector (11)—asserted as a *matrix*, and explicitly *not* the identity (idempotence alone would admit it):

```
def test_commit_projector_value():
    K = Matrix([[0, 0], [1, 0], [2, 0], [0, 1]])      # kernel basis e1+2e2, i
    P = K * (K.T * K).inv() * K.T                      # exact orthogonal projector onto ker(L)
    assert P == Matrix([[0, 0,     0,     0],
                        [0, Q(1,5), Q(2,5), 0],
                        [0, Q(2,5), Q(4,5), 0],
                        [0, 0,     0,     1]])
    assert P * P == P                                  # idempotent
    assert P != eye(4)                                 # ... and not the trivial projector
```

Second, the witness digest of Example 2.5, recomputed from the full real body and asserted to the exact 16-hex value (the fix from this revision—no length/flip proxy):

```
def test_witness_digest_exact():
    body = {"event": "basis_growth", "index": 0, "min_poly": [1, 0, -24],
            "coords": ["-5", "0", "1", "0"], "snap": "exact",
            "num_seeds": 3, "streak": 4, "prev_hash": "genesis"}
    canonical = json.dumps(body, sort_keys=True, separators=(",", ":"))
    digest = hashlib.sha256(("genesis" + canonical).encode()).hexdigest()[:16]
    assert digest == "31f1f1e05ac9a35a"
```

Third, the "characteristic polynomial is insufficient" witness (Remark 3.3): equal characteristic polynomials, different minimal polynomials, hence not similar:

```
def test_charpoly_insufficient_phi_witness():
    A = sp.diag(companion_high([1,-1,-1]), companion_high([1,-1,-1]))   # phi (+) phi
    B = companion_high([1,-2,-1,2,1])                                   # C((x^2-x-1)^2)
    assert sp.expand(A.charpoly(x).as_expr() - B.charpoly(x).as_expr()) == 0   # SAME charpoly
    I4 = eye(4)
    assert A*A - A - I4 == zeros(4,4)        # A satisfies x^2-x-1  (minpoly x^2-x-1)
    assert B*B - B - I4 != zeros(4,4)        # B does NOT -> minpoly (x^2-x-1)^2 -> not similar
```

8.3 Authoritative suite counts

All green at the pinned heads (run with the `py` launcher on Windows; §A):

Suite	Result	Pinned at
L00M training suite (`py -m pytest training`)	127 passed (137 with this revision)	L00M `038ee27`
Full L00M repository suite (`py -m pytest`)	544 passed (574 with this revision)	L00M `038ee27`
ZFP zero-free-parameter gate	PASS:74 FAIL:0	Plate-Matrices `5e78d67`
kira-language suite (`py -m pytest`)	121 passed	kira-language `44829ff` (
Vector-substrate paper probe	13 passed	L00M `2785980`
Companion paper probe (this paper)	20 passed	§A
Calibration tests (`test_calibration.py`)	4 passed	L00M `1366d1c`

8.4 The claim→test→hash map

Table 5 maps each displayed claim of §§2–7 to the machine-checked test that proves it and the commit that pins it.

8.5 Computational aspects

Every operation on both faces is exact rational (or integer) arithmetic, trading floating speed for reproducibility. Forming a trace-form Gram is $O(n^3)$ rational multiplications (or $O(n^2)$ trace evaluations given a multiplication table); the projector solve $B(B^\top GB)^{-1}B^\top G$ is an $O(n^3)$ exact linear solve, and in streaming use one appends a column and updates the small factorisation of $B^\top GB$ rather than refactoring. The seed-naming bridge is a Smith normal form of $xI - \rho(\alpha)$ over $\mathbb{Q}[x]$; the admissibility gate is $O(n)$ integer comparisons; the non-disjoint factorisation is bounded Kronecker, capped at degree 12. On the language face the kernel and projector are fixed 4×4 rational constants computed once, and lexicon membership is a content-hash lookup. The single inexact step anywhere is the threshold comparison against $\log \mathrm{M}$, evaluated by certified interval arithmetic (Example 4.4) while the gain side stays an exact rational; rational coefficient growth is the practical caveat (Remark 4.5), bounded by the same capacity gate. Concretely, growing the forced basis in $\mathbb{Q}(\sqrt{2}+\sqrt{3})$ from $\{1\}$ through $\{1, \theta\}$ to $\{1, \theta, 2\sqrt{6}\}$ keeps the exact projector entries small—largest numerator 49, largest denominator 5 (probe-measured)—because the degree/height budget caps the seeds; it is precisely an *unbounded* working degree that would let the numerators and denominators blow up, which the gate forbids. None of this alters a decision—it only prices exactness.

9 The honesty ledger

We restate the provable/computed/conjectural split in the substrate's format.

10 Open problems

(O1) **The exchange rate λ *(resolved: the identity* $\lambda = 2c$*).*** The information price of one basis direction is no longer posited. Reading the trace form as the Fisher metric G/c turns the growth rule into the two–part MDL code, whose exchange rate is the identity $\lambda = 2c$ (Theorem 4.6); the probe's $\lambda = 2$ is its $c = 1$ (Gaussian) instance and the degree–aware floor its $c = n$ instance (Remark 4.7). This collapses (O1) into (O4): the exchange rate and the conformal constant are a single degree of freedom, related by $\lambda = 2c$, so "derive λ" becomes "declare c," addressed next.

(O2) **The reciprocal-seed floor.** A uniform positive Mahler floor is a theorem only for non-reciprocal seeds (Smyth); the reciprocal case—Salem numbers and units, which arise naturally as seeds—is the unsolved core of Lehmer's problem, and the gate is honest that it has no proven floor there.

(O3) **The non-disjoint compositum beyond the bound.** The construction of §3.3 is exact up to the Kronecker degree cap; a Zassenhaus/p-adic factorisation over number fields would remove the bound (Remark 3.11).

(O4) **The conformal constant c *(reframed: a declared scale with a critical–point–native canonicalization).*** With $\lambda = 2c$ the only residual freedom is c, and by Čencov's theorem it is irreducible *within* information geometry—no invariance argument fixes the scale [16] (Remark 4.8). Three readings of the field's own structure name three values (Table 2): $c = 1$ (Jeffreys information volume $\sqrt{|d_K|}$), $c = n$ (degree–invariant per–conjugate significance), and the critical–point value $c = \sqrt{1 + 4C}/(2C)$ selected by the self–action eigenframe at a gate (Definition 4.10; golden gate $c = \sqrt{5}/2$, $\lambda = \sqrt{5}$). The third is native to the framework—the eigenframe's own scale rather than an imported prior—but, per Čencov, it is a principled *choice*, not a derivation. What remains genuinely open is therefore narrow and named: not the *value* of λ (the identity $\lambda = 2c$ settles that) but *which* declared conformal model an application adopts. The shipped substrate adopts $c = 1$, and the shipped seeds' decisions are invariant across $c \in [1, n]$ (Remark 4.11).

(O5) **The live language wire.** Wiring the language shell into a host service additively (the B1 step) is engineering, not mathematics, and is out of scope: this paper covers the prepared, machine-verified surface, not its deployment.

Remark 10.1 (Scope and limitations). The exact layer of both faces—capture, residual, projector, the seed-naming bridge, admissibility, the return operator and its kernel, the lexicon dedup, the witness chains—is decided over $\mathbb{Q}/\mathbb{Z}$ and is proved or machine-verified. The non-disjoint construction is exact within a declared degree bound. The thresholding layer is not exact ($\log \mathrm{M}$ is transcendental) but no longer carries a posited rate: the exchange rate is the identity $\lambda = 2c$ (Theorem 4.6), and the only residual modeling freedom is the conformal constant c—a declared scale, irreducible by Čencov's theorem (Remark 4.8), with a critical–point–native canonicalization (Definition 4.10). The reciprocal floor remains open. The information-geometry reading is conformal with this declared constant and requires $1 \in \mathrm{col}(B)$ (and, for positive–definiteness, that the field be totally real, Remark 4.12). The framework yields finiteness of the admissible seed set but no sample-complexity bound. None of these gaps touches the exact core; each is flagged where it arises and collected here.

A Reproducibility

Every numerical claim is reproduced by a machine-checked probe or an existing suite test; we list the commands and the provenance.

Suites (Windows, `py` launcher, `PYTHONIOENCODING=utf-8`).

- **Learning, cross-field, capacity, loop:** `cd LOOM; py -m pytest training -q` $\Rightarrow$ 127 passed (HEAD `038ee27`). Per cluster: residual learner 18 (`test_residual_learner.py`, `218c02e`); bridge 7 (`test_coords_to_minpoly.py`, `c1db881`); compositum 31 (`test_compositum.py`, `test_compositum_n` `test_field_extension.py`, `3e399f1/75524b3`); capacity/Fisher/threshold 29 (`test_capacity.py`, `test_capacity_threshold.py`, `test_a3p2_fisher.py`, `test_a3p2b_threshold.py`, `c8fdb8a/d526098/c5605` auto-loop 28 (`test_anomaly_detector.py`, `test_field_growing_learner.py`, `test_detector_driven_gain.` `038ee27/6ad20eb/ef3b86d`). With this revision the training suite is 137 (the 4-test `test_calibration.py` for the concrete variance calibration of Remark 5.2, commit `1366d1c`, plus the 6-test `test_a3p2c_lambda_confo` for the $\lambda = 2c$ closure of §4.3).
- **Full repository:** `py -m pytest -q` $\Rightarrow$ 544 passed (HEAD `038ee27`, before this work); 574 with this revision's additions (the 20-test companion probe, the 4-test `test_calibration.py`, and the 6-test `test_a3p2c_lambda_conformal.py`).

- **ZFP gate:** `cd Plate-Matrices; py zfp-pipeline/zfp_master_verify.py` ⇒ PASS:74 FAIL:0 (HEAD `5e78d67`).
- **Language:** `cd kira-language; py -m pytest -q` ⇒ 121 passed (HEAD `44829ff`; tag `b1-ready` = `a5892c7`). Holdings `f925e50/6cd3800`; acquisition `4817b62`; lexicon `d1312ed`; store `663e00e`; dispatch `f06ba5d`.

Companion probe. The additive, model-layer-only probe `paper/test_residual_return_claims.py` (sympy + stdlib, importing no engine) recomputes every *displayed number* of this paper: the power-basis Gram (2); the residual norms 96, 56, 10; the seed minimal polynomials (x^2-24, x^2-7, x^2-x-1); the $\varphi \oplus \varphi$ and Jordan invariant-factor witnesses; the non-disjoint degree-6 factorisation $x^6-25x^4+91x^2-75=(x^2-3)(x^4-22x^2+25)$ and the reconstruction $\beta = -\frac{7}{20}\theta + \frac{1}{20}\theta^3$; the Fisher matrices of Example 4.2 and $G = n\mathcal{I}$ on trace-zero; the Smyth bracket and the floors $0.5624, 2.2496, 4.4992$; and the language kernel basis, the commit projector (11), $\mathcal{L}(H(0,-1,1,0)) = H(-3,0,0,0)$, the words word($\pm i$), and the content-hash identity; and—added in this revision—the witness digest `31f1f1e05ac9a35a` recomputed from the *full* `_witness_growth` body (`test_witness_digest_exact`) and the certified-GROW enclosures verified by `mpmath.iv` interval arithmetic (`test_certified_grow_enclosures`, 20 tests total). Run: `cd L00M/paper; py -m pytest test_residual_return_claims.py -q`.

Derivation checks (this revision, §4.3). The symbolic results of the exchange–rate section are reproduced by exact computation (`sympy` + `mpmath`, importing no engine), additive supplements to the suite above. The identity $\lambda = 2c$ is checked by solving the MDL balance $D_{\mathrm{KL}} = \frac{1}{2c}\|\mathbf{r}\|_G^2 \geq \log \mathrm{M}$ for the threshold and reading off the rate ($\lambda = 2c$, $\lambda|_{c=1} = 2$, $\lambda|_{c=n} = 2n$); the floor unification of Remark 4.7 recomputes $2c \log \mu_S$ at $c \in \{1, n\}$ to $0.5624, 2.2496, 4.4992$ from μ_S, the real root of $x^3 - x - 1$; the self–action spectrum of Proposition 4.9 is verified by forming the 4×4 adjoint matrix $[R_C, \cdot]$ and confirming its eigenvalues $\{0, 0, \pm\sqrt{1+4C}\}$ and kernel span$\{I, R_C\}$, with the golden case $\{-\sqrt{5}, 0, +\sqrt{5}\}$ and R_1 conjugate to the keystone; and the canonicalization ladder of Table 2 evaluates $c = \sqrt{1+4C}/(2C)$ on the gates $C \in \{\frac{1}{4}, \frac{1}{2}, 1\}$ to $\{2\sqrt{2}, \sqrt{3}, \sqrt{5}/2\}$ with gap $\sqrt{1+4C} \in \{\sqrt{2}, \sqrt{3}, \sqrt{5}\}$. Every conformal/companion term is exact ($\mathbb{Q}/\mathbb{Z}$ or $\mathbb{Q}(\sqrt{5})$); only the transcendental $\log \mu_S$ uses a certified interval bracket.

Provenance. The dynamic learning/loop results correspond to the training cluster at L00M HEAD `038ee27`; the Fisher and threshold probes to `d526098` and `c5605d7` (shared with the substrate paper); the non-disjoint construction to `3e399f1`; the language layer to the kira-language readiness tag `b1-ready` (`a5892c7`, HEAD `44829ff`). All displayed values match these probes.

References

[1] L00M Vector Substrate Project, *The Vector Substrate: Number Fields as Exact Learning Geometry*, companion manuscript, 2026.

[2] C. J. Smyth, *On the product of the conjugates outside the unit circle of an algebraic integer*, Bull. London Math. Soc. **3** (1971), 169–175.

[3] D. H. Lehmer, *Factorization of certain cyclotomic functions*, Ann. of Math. **34** (1933), 461–479.

[4] E. Dobrowolski, *On a question of Lehmer and the number of irreducible factors of a polynomial*, Acta Arith. **34** (1979), 391–401.

[5] K. Mahler, *An application of Jensen's formula to polynomials*, Mathematika **7** (1960), 98–100.

[6] D. G. Northcott, *An inequality in the theory of arithmetic on algebraic varieties*, Proc. Cambridge Philos. Soc. **45** (1949), 502–509.

[7] E. Bombieri, W. Gubler, *Heights in Diophantine Geometry*, Cambridge University Press, 2006.

[8] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften **322**, Springer, 1999.

[9] D. A. Marcus, *Number Fields*, Springer, 1977.

[10] D. S. Dummit, R. M. Foote, *Abstract Algebra*, 3rd ed., Wiley, 2004.

[11] H. Cohen, *A Course in Computational Algebraic Number Theory*, Graduate Texts in Mathematics **138**, Springer, 1993.

[12] P. Lounesto, *Clifford Algebras and Spinors*, 2nd ed., London Math. Soc. Lecture Note Series **286**, Cambridge University Press, 2001.

[13] S. Amari, *Information Geometry and Its Applications*, Applied Mathematical Sciences **194**, Springer, 2016.

[14] S. Amari, H. Nagaoka, *Methods of Information Geometry*, Translations of Mathematical Monographs **191**, AMS, 2000.

[15] D. Lind, K. Schmidt, T. Ward, *Mahler measure and entropy for commuting automorphisms of compact groups*, Invent. Math. **101** (1990), 593–629.

[16] N. N. Čencov, *Statistical Decision Rules and Optimal Inference*, Translations of Mathematical Monographs **53**, AMS, 1982.

[17] J. Rissanen, *Modeling by shortest data description*, Automatica **14** (1978), 465–471.

[18] H. Jeffreys, *An invariant form for the prior probability in estimation problems*, Proc. Roy. Soc. London Ser. A **186** (1946), 453–461.

Algorithm 1: The exact streaming learner (§2). Every quantity is $\mathbb{Q}/\mathbb{Z}$ and every branch is an exact comparison; floats never enter a decision. *confirm* is the only line that mutates B.

Data: forced basis B, projector P; Welford state $(\text{mean}, M2, n)$; streak s; persistence N, tolerance ε

function observe($\mathbf{x}$)**:**

    $\mathbf{r} \leftarrow \mathbf{x} - P\mathbf{x}$; $\rho \leftarrow \mathbf{r}^{\mathsf{T}} G\, \mathbf{r}$ `// exact residual and its` G`-norm`
    if $\rho = 0$ **then** reset(); **return**   `// captured: no novelty`

    if $n = 0$ **then**
        welford($\mathbf{r}$); $s \leftarrow 1$
    else
        $c \leftarrow \text{mean}^{\mathsf{T}} G\, \text{mean}$
        if $c = 0$ **then** reset(); welford($\mathbf{r}$); $s \leftarrow 1$; **return**
        $\kappa \leftarrow (\mathbf{r}^{\mathsf{T}} G\, \text{mean})/c$; $\text{dev}^2 \leftarrow (\mathbf{r} - \kappa\, \text{mean})^{\mathsf{T}} G\, (\mathbf{r} - \kappa\, \text{mean})$
        if $\text{dev}^2 \le \varepsilon^2 \rho$ **then**
            welford($\mathbf{r}$); $s \leftarrow s + 1$ `// direction aligned`
        else reset(); welford($\mathbf{r}$); $s \leftarrow 1$ `// direction changed`

function welford($\mathbf{r}$)**:**

    $n \leftarrow n + 1$
    for *each coordinate* i **do**
        $\delta \leftarrow \mathbf{r}_i - \text{mean}_i$; $\text{mean}_i \leftarrow \text{mean}_i + \delta/n$; $M2_i \leftarrow M2_i + \delta\,(\mathbf{r}_i - \text{mean}_i)$

function propose()**:**

    if $s < N$ **or not** calibrated() **or** mean $= \mathbf{0}$ **then return** NONE
    $(\text{coords}, m, \text{snap}) \leftarrow \text{seedFromCentroid}(\text{mean})$ `// exact bridge; else snap (Rem. 3.4)`
    if $m =$ *NONE* **or** residual(*coords*) $= 0$ **then return** NONE
    if capacity(m, $\text{mean}^{\mathsf{T}} G\, \text{mean}$, *budget*) $\neq$ *GROW* **then return** NONE
    return SeedProposal(m, coords, s, snap)   `// pure:` B `unchanged`

function confirm(p)**:**

    $m \leftarrow$ minpoly(p.coords)   `// re-validate algebraic integer (G10)`
    $B' \leftarrow [\,B \mid p.\text{coords}\,]$; $P' \leftarrow B'(B'^{\top} G B')^{-1} B'^{\top} G$
    if $P' =$ *NO_PROJECTION* **then raise** (degenerate growth refused)
    $B \leftarrow B'$; $P \leftarrow P'$ `// ◀ the sole mutation of` B
    witness(p, m)   `// append SHA-256 link (3)`
    ; reset()

Canonicalization	principle	c	status
Model 1	Jeffreys volume $\sqrt{\det G} = \sqrt{\lvert d_K \rvert}$	1	declared [18]
Model 2	degree–invariant per–conjugate significance	n	declared
Critical point	self–action eigenframe scale (gap $\sqrt{1+4C}$)	$\sqrt{1+4C}/(2C)$	declared; framework–native

Table 2: The conformal constant c under three canonicalizations, all giving $\lambda = 2c$ by Theorem 4.6. By Čencov's theorem (Remark 4.8) none is forced by invariance; the third is selected by the geometry of a critical point (Definition 4.10), and at the golden gate gives $c = \sqrt{5}/2$, $\lambda = \sqrt{5}$. On the ladder gates $C \in \{\frac{1}{4}, \frac{1}{2}, 1\}$ the gap is $\sqrt{1+4C} \in \{\sqrt{2}, \sqrt{3}, \sqrt{5}\}$, the three irrationals $\sqrt{\mathrm{M}(\sqrt{2})}, \sqrt{\mathrm{M}(\sqrt{3})}, \sqrt{\mathrm{M}(\sqrt{5})}$.

Algorithm 2: The automatic detector-driven loop (§5, Theorem 5.4). The gain is *measured*, not supplied; the gate judges the *true* compositum degree $[K:\mathbb{Q}]\,e'$ (not the tensor degree); *confirm* is the sole field-growth mutation; re-home returns the residual to 0 in $\mathbb{Q}(\theta')$.

Input: ambient $K = \mathbb{Q}(\alpha)$, forced basis B with $1 \in \mathrm{col}(B)$, an out-of-field stream; budget
for *each observation* $\mathbf{x}$ **do**
  $s \leftarrow \|\mathbf{x} - P\mathbf{x}\|_G^2$ `// detect: exact residual score`
  if $s > 0$ **and** calibrated() **then**
    $g \leftarrow$ measuredFieldResidual($\mathbf{x}$) `// auto-gain = the MEASURED out-of-field residual`
    $(\beta, m_\beta) \leftarrow$ candidateGenerator($\mathbf{x}$)
    factor m_β over K; $e' \leftarrow$ degree of the K-factor satisfied by β `// effective degree`
    if capacity$\big(\cdot, g,$ budget on $[K:\mathbb{Q}]\,e'\big) =$ GROW **then**
      $\theta' \leftarrow$ primitive element of $K(\beta)$; $m_{\theta'} \leftarrow$ its minpoly `// Thm. 3.9`
      re-home each column of B into the power basis of θ' (clear denominators, Lem. 5.5)
      confirm `// ◀ sole field-growth mutation; afterwards` $\|\mathbf{x} - P\mathbf{x}\|_G^2 = 0$
    else REJECT/STOP: surface a proposal, do not grow

Jurisdiction	Wired?	Count in law bank	Meaning
THEOREM	yes	20	proved (or exact-verified) fact
COMPUTED	yes	5	derived/learned (e.g. a lexicon entry, a reading)
INTERPRETIVE	no	2	glossed reading—returned only on explicit request, tagged
FALSE__AS__STATED	no	0	defined for completeness; unused
wired total	—	**25**	what crosses by default $(20+5)$
bank total	—	**27**	all statements $(20+5+2)$

Table 3: The jurisdiction firewall of the language layer. WIRED__JURISDICTIONS = $\{$THEOREM, COMPUTED$\}$; the default surface returns the 25 wired statements, the full 27 only on explicit request with each non-wired statement tagged. Machine-verified (`test_firewall`, `test_math_preserved`; commit `a5892c7`).

Role	Learning (number field K)	Language ($\mathrm{Cl}(2,0) \cong M_2(\mathbb{R})$)
loss / residual	$\mathbf{r} = \mathbf{x} - P\mathbf{x}$ (trace form G)	$\mathcal{L}(X) = RX + XR - X$
captured $\iff$	$\|\mathbf{r}\|_G^2 = 0$ (exact, over $\mathbb{Q}$)	$\mathcal{L}(X) = \mathbf{0}$ (exact, over $\mathbb{Q}$)
landing space	$\mathrm{col}(B) \subseteq K$	$\ker \mathcal{L} = \mathrm{span}\{e_1 + 2e_2,\, i\}$
model state	forced basis B (growing)	lexicon $\subseteq \ker \mathcal{L}$ (growing)
sole mutator	*confirm* (adjoin seed)	*commit* / lexicon *add*
generalisation	same residual direction captured once	same exact residue $\Rightarrow$ one entry
witness	SHA-256 chain (3)	*the same* SHA-256 chain
exactness gate	$\mathbb{Q}/\mathbb{Z}$ decision; float refused	$\mathbb{Q}$ decision; float refused

Table 4: The residual-return correspondence. The two columns are one mechanism in two carriers; the witness chain is literally the same construction in both.

Claim (where)	Test	Commit
Exact capture, zero tolerance; float refused (Prop. 2.1)	`test_a_captured_input_...`; `test_g8_exact_fraction_core_floats_rejected`	`218c02e`
confirm is the sole mutator; *propose* idempotent (Prot. 2.2)	`test_g2_confirm_is_the_only_mutator`	`218c02e`
Exactly one growth; residual returns to 0 (Thm. 2.3, Ex. 2.4)	`test_b_persistent_offaxis_...`; `test_c_after_confirm_...`	`218c02e`
Witness digest `31f1f1e05ac9a35a` from the FULL body (Ex. 2.5)	`test_witness_digest_exact` (probe, exact value); `test_g5_...` (tamper only)	probe; `218c02e`
Bridge m_α = largest invariant factor (Thm. 3.1, Ex. 3.2)	`test_coords_to_minpoly.py` (7 tests)	`c1db881`
Exact centroid; nearest-integer snap fallback (Rem. 3.4)	`test_coords_to_minpoly.py`; `_seed_from_centroid` path	`c1db881`, `218c02e`
$\varphi \oplus \varphi$ / Jordan: charpoly insufficient (Rem. 3.3)	`test_similarity_caveat_phi_dsum_phi`; `invariant_factors` demo	`c1db881`
Disjoint Kronecker Gram; $\sqrt{7}$ growth (Prop. 3.5, Ex. 3.6)	`test_p2b_sqrt7_...`; `test_tensor_gram_is_kronecker_...`	`3e399f1`, `75524b3`
Non-disjoint $m_\theta = x^4 - 22x^2 + 25$; factor selection (Thm. 3.9, Ex. 3.10)	`test_build_compositum_witness`; `test_p2c_witness_grows_true_compositum_not_tensor`; `test_factor_over_Q_correctness`	`3e399f1`
Northcott gate exact; Landau certificate (Ex. 4.1)	`test_capacity.py` (incl. `...float_input_rejected_at_gate`)	`c8fdb8a`
Exact Fisher matrices; $G = n\mathcal{I}$ on trace-zero (Ex. 4.2)	`test_a3p2_fisher.py`	`d526098`
Smyth floor; degree-aware floor; reciprocity caveat (Rem. 4.3)	`test_capacity_threshold.py`; `test_a3p2b_threshold.py`	`c8fdb8a`, `c5605d7`
Detector: propose, never auto-grow; calibration (Prop. 5.1)	`test_a7_*` (in-dist, calibration, no-auto-grow, witness)	`6ad20eb`, `038ee27`
Concrete variance calibration; persists $\neq$ calibrated (Rem. 5.2)	`test_calibration.py` (stable calibrates; volatile blocks)	`1366d1c`
Auto-loop returns residual to 0; effective degree (Thm. 5.4)	`test_e2e_...`; `test_seam_...`; `test_effective_degree_...`	`038ee27`, `ef3b86d`
Re-home projector-invariant; $1 \in \mathrm{col}(B)$ preserved (Lem. 5.5, Rem. 5.6)	`test_rehome_preserves_subspace_...`; `test_one_in_basis_preserved_...`	`ef3b86d`
Exact carrier; readings zero-tolerance (§6)	`test_holding.py`; `test_carrier_axioms_and_cocycle_exact`	`f925e50`, `6cd3800`
$\mathcal{L}$ oracle-checked; $R \notin \ker \mathcal{L}$ (Prop. 6.2)	`test_L_matches_matrix_oracle_exact`; `test_keystone_R_not_in_kernel_exact`	`4817b62`
$\ker \mathcal{L}$ exact, dim 2; the disproof (Thm. 6.3, Rem. 6.4)	`test_kernel_exact_dim_and_basis`; `test_corrected_handoff_hint_is_not_in_kernel`	`4817b62`
Commit projector value (11); generalisation; ids (Thm. 6.5)	`test_projector_exact_matrix_value`; `test_generalization_...`; `test_distinct_residues_...`; `test_ids_are_deterministic_...`	`4817b62`, `d1312ed`
Firewall; one-way; 121 green (§6)	`test_firewall`; `test_one_way_and_live_phi`; `test_b1_readiness.py`	`a5892c7`
φ-keystone is one object (Prop. 7.1)	`test_three_route_closure_on_phi_keystone`; `test_keystone_equals_loom_bridge_phi_cl`	`371e09`, `a5892c7`

Table 5: The claim→test→hash map: every displayed claim of this paper, the named machine-checked test that proves it, and the commit that pins it. Suite-level counts are in the text above; the companion probe (§A) re-derives every displayed *number*.

Statement	Status	Where
Exact capture criterion $\mathbf{r} = 0 \iff$ captured	Cited (proved) + probe	[1], Prop 2.1
Propose-for-confirm; sole mutator; witness chain	This work (design) + probe	Prot 2.2, Thm 2.3
Bridge m_α = largest invariant factor	Cited (proved) + probe	[1], Thm 3.1
Charpoly insufficient ($\varphi \oplus \varphi$, Jordan)	Classical + probe	Rem 3.3
Disjoint compositum $G = G_K \otimes G_L$	Cited (proved) + probe	[1], Prop 3.5
Non-disjoint compositum (bounded), $m_\theta = x^4 - 22x^2 + 25$	This work (exact-computed)	Thm 3.9, Ex 3.10
Northcott gate exact; Landau certificate	Cited (proved) + probe	[1], Ex 4.1
$G = n\,\mathcal{I}_{\exp}$ on trace-zero ($1 \in \operatorname{col} B$)	Cited (this work in [1]) + probe	[1], Ex 4.2
Smyth floor μ_S (non-reciprocal only)	Cited (theorem)	Rem 4.3
Reciprocal-seed floor (Salem, units)	Open (no uniform floor)	Rem 4.3
Automatic loop returns residual to 0	This work (design) + probe	Thm 5.4
Concrete variance calibration (Welford-$M2$ gate; default permissive)	This work (design) + probe	Rem 5.2
Re-home projector-invariance; $1 \in \operatorname{col} B$ kept	This work (proved) + probe	Lem 5.5, Rem 5.6
$\ker \mathcal{L}$ exact, two-dimensional	This work (proved) + probe	Thm 6.3
Commit projector value; generalisation by return-to-zero	This work (proved) + probe	Thm 6.5
φ-keystone is one object (three routes)	This work (exact-computed)	Prop 7.1
Firewall: only THEOREM/COMPUTED cross	This work (design) + probe	§6
Exchange rate identity $\lambda = 2c$	This work (derived)	Thm 4.6
Self–action spectrum $\{-\sqrt{5}, 0, +\sqrt{5}\}$; gap $\sqrt{1+4C}$	This work (proved) + computed	Prop 4.9
Conformal constant c (Čencov: free up to scale)	Declared; named $c \in \{1, n, \sqrt{1+4C}/(2C)\}$	Rem 4.8, Def 4.10
Non-disjoint compositum beyond the degree cap	Open	Rem 3.11

Table 6: Honesty ledger. *Cited (proved)*: proved in [1], re-verified here by probe. *This work*: assembled, proved, or constructed in this paper. *Classical*: standard, re-verified. *Exact-computed*: a certified exact computation (the non-disjoint witness, the keystone identity). *Cited (theorem)*: Smyth's floor, unconditional only for non-reciprocal seeds. *Open/Conjectural/Model-dependent*: the genuinely unsettled parts. Every "+ probe" row is reproduced by a machine-checked computation (§8, §A).